CMMS for Sree Murugan Tile Works — Software Requirements Specification & Design Document

Version 0.4 (Draft for review) — Features/Design Phase

Prepared for Lakshmi Anand K (Software Architect)

12 October 2026

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Document control

Item	Value
Document	Software Requirements Specification (SRS) & High-Level Design — in-house CMMS
Version	0.4 — draft for review
Date	12 Oct 2026 (IST)
Owner	Lakshmi Anand K, Software Architect
Target stack	Vue 3 (frontend, PWA), Flask / Python (REST API), MariaDB; integrates with ERPNext v16 (Frappe)
Hosting	Self-hosted on-premise; deployed with the existing PyCharm → Ansible pipeline
Status	Features / design phase — not yet baselined

Change log

Version	Date	Author	Changes
0.1	12 Oct 2026	Architect (with research assistant)	First draft: competitive analysis, FRs, NFRs, use cases, user stories, architecture, roadmap, open questions. Updated with confirmed sites (Main Plant, Ancillary Plant, Campus) and ~500 machines.
0.2	12 Oct 2026	Architect (with research assistant)	Added the user's answers to open questions. **(1) Devices:** technicians are phone-first (mobile PWA is their primary UX); engineers use mobile and desktop, so rich desktop workspaces were added (FR-DSK-xxx). **(2) Hosting:** self-hosted on-premise with the PyCharm → Ansible pipeline. Added deployment architecture (§8.4), Ansible roles, HTTPS/certificate strategy for the PWA, inter-site VPN access, and updated backup/deployment NFRs. **(3) Greenfield:** no history to migrate. Bulk import templates for assets, spares and BOMs kept and expanded (FR-AST-008, FR-AST-014, FR-SPR-020). **(4) ERPNext** is used for POs and inventory; Tally integration is out of scope. Added new **Section 9: Spare parts & ERPNext integration** (options A/B/C, recommendation, system-of-record split, sync patterns, warehouse mapping, reservations/kitting, returns, repairables, sequence diagrams). FR-INV/FR-PRC replaced by FR-SPR-xxx and FR-ERP-xxx. Added UC-13…UC-18 and US-43…US-54. Updated ERD, API, roadmap and sources. Answered items removed from assumptions and open questions. Sections renumbered: Roadmap → §10, Open questions → §11, Sources → §12.
0.3	12 Oct 2026	Architect (with research assistant)	Added the user's second set of answers. **(1) ERPNext v16** confirmed, with a test instance for development; v16 specifics verified and cited (stock reservation scope, Item Reorder MR types, Bin fields, Stock Entry purposes, webhook events/timeout/no retry count). Added a **read-mostly master pull** (UOM, Item Group, Warehouse, Item, Bin, Supplier, Serial No) and contract tests (FR-ERP-006, FR-ERP-012, US-61). **(2) Stores:** warehouse mapping revised to **one master store plus a few sub-stores** (not per-site uniform), with central Kitting/Quarantine, workshop WIP, vendor and transit warehouses; cross-store availability; inter-store transfers via ERPNext Material Transfer, now in the MVP (§9.4, FR-SPR-003/005/018/022, UC-19, US-57). **(3) Repair locations:** WO `repair_location` (In-situ / Mechanical shop / Electrical shop / External), workshops and repair jobs (new §4.20 FR-WSH-xxx, FR-WO-014/015), workshop and external-repair flows, and external-repair options R1/R2/R3 with R1 recommended (§9.5, §9.7.5–9.7.6, FR-SPR-017/023/024, UC-17,

Version	Date	Author	Changes
			UC-20, US-58–60). **(4) Storekeeper issues; technicians request only;** Option A confirmed; the CMMS holds no stock ledger (FR-SPR-007/008/009, US-55; self-issue removed). **(5) Admin approves** maintenance MRs, POs (in ERPNext), external repairs, reorder pushes and scrap in a single step (new FR-APR-001…004, UC-21, US-56). Roles, ERD, API, diagrams (integration, issue, shortage, transfer, kitting, workshop repair, external repair), roadmap, open questions and sources (S79–S88) updated. Section 9 renumbered: flows §9.7, FRs §9.8, use cases §9.9, stories §9.10.
0.4	12 Oct 2026	Architect (with research assistant)	**(1) Exactly one Admin per software instance** (one single approver across all sites, no per-site admins, no second approver or value threshold). The leave-cover delegate stays as an optional Should (FR-APR-003); per-site/threshold questions removed. **(2) Stores issue mapping:** a storekeeper issue against a WO posts an ERPNext Stock Entry with purpose **Material Issue** (consumption), carrying the WO reference, cost centre and expense account. Inter-store, kitting and repair moves use **Material Transfer**. Added §9.4.1 explaining the difference from the user's "material issue voucher", plus the alternative mapping if stores today book issues as a Material Transfer to a consumption/department warehouse (new open question 1). Updated FR-SPR-008, FR-ERP-001, US-44, sequence 9.7.1 and the open questions.

How to read this document. Statements about commercial products and ERPNext are limited to what the cited sources (Section 12) support at the time of research (Oct 2026). Vendor tiers and prices change; re-check them before any decision that depends on them. Anything about Sree Murugan Tile Works' plants, people, processes or volumes that the user has **not** confirmed is tagged **[ASSUMPTION — validate]**.

Confirmed facts (from the user):

- **Organisation:** Sree Murugan Tile Works, manufacturing, India.
- **Sites:** three sites: **(1) Main Plant, (2) Ancillary Plant, (3) a smaller Campus.**
- **Equipment:** about **500 machines and equipment in total** across the three sites.
- **Stack:** Vue (frontend), Flask/Python (backend), MariaDB (database).
- **Goals:** encourage maintenance best practices, standardisation and visibility.
- **Devices:** maintenance **technicians use phones** for most work, so the mobile-first PWA is their primary UX. **Engineers use both mobile and desktop**, so rich desktop views are needed for planning, analytics and configuration.
- **Hosting: self-hosted on-premise**. The existing CI/CD is **PyCharm → Ansible** deployment.
- **Data: greenfield**. There is no existing maintenance history, spares list or PM schedule to migrate.
- **ERP: ERPNext v16** is used for purchase orders and inventory, and **inventory stays managed in ERPNext** (the CMMS holds no stock ledger). A **test ERPNext instance** will be provided for development. The CMMS syncs item master, UOMs, item groups, warehouses and stock levels **from** ERPNext (read-mostly). **Tally** is used for accounts; Tally integration is **out of scope** for now.
- **Stores: one master store plus a few sub-stores** (not necessarily one per site).
- **Repairs** are done **in-situ** (on the machine), in the **Mechanical shop**, in the **Electrical shop**, or **externally** (vendor/subcontract).
- **Stores control: storekeepers issue parts; technicians only request.**
- **Approvals:** each instance of the software has **exactly one Admin**, the single approver across all sites for maintenance Material Requests and POs.

1. Executive summary

Sree Murugan Tile Works is building an in-house Computerised Maintenance Management System (CMMS) for its three sites (Main Plant, Ancillary Plant, Campus). Together they have about 500 machines and equipment. This document pulls together research on eight commercial CMMS/EAM products (MaintainX, UpKeep, Fiix, Limble, eMaint, IBM Maximo, SAP PM/S/4HANA Asset Management, and Infor EAM, now HxGN EAM) and two open-source options (Odoo Maintenance, openMAINT). It also covers the relevant standards (ISO 55000, ISO 14224, EN 13306, SMRP metrics, OEE, TPM/5S). From these it builds a requirements baseline and a high-level design.

Key findings

1. **Every product shares the same core:** an asset register with a parent/child hierarchy, work requests that become work orders, recurring preventive maintenance (PM), checklists/procedures, parts inventory, and reporting. Products differ in mobile UX, how deep inventory and purchasing go, condition-based/IoT triggers, multi-site control, and analytics.
2. **Commercial vendors put the "standardisation" features in higher tiers.** Examples:
    - MaintainX Premium: meter-based PM, parts inventory, purchase orders, API access.
    - Fiix Professional/Enterprise: multi-site, failure codes, audit trail.
    - Limble Enterprise: multi-location, custom roles/approval workflows, ERP/IoT integration.
    - UpKeep Professional and eMaint Professional: offline mobile.

    An in-house build can include all of these from day one, with no per-seat cost.
3. **Free requesters are the norm.** MaintainX, UpKeep and Limble all let anyone submit requests without a paid seat, often through a QR code or web link. This matters for visibility: operators must be able to raise a request in seconds.
4. **Offline-capable, mobile-first work execution is standard.** All major vendors offer offline work orders with sync: MaintainX, Limble, UpKeep Professional, eMaint Professional, Maximo Mobile, SAP Service and Asset Manager, and HxGN EAM Mobile.
5. **Enterprise EAMs separate the problem report from the planned work.** SAP PM keeps the *notification* (the problem report, with catalog codes) apart from the *order* (the planned work). Its maintenance plans can generate either one. This is a good pattern for collecting structured failure data.
6. **The standards give us a ready-made data model:**
    - ISO 14224: equipment taxonomy (the equipment unit and the items below it) and failure data (failure mode, mechanism, cause, detection method).
    - EN 13306: maintenance types (preventive, corrective, condition-based, predictive).
    - SMRP-style KPIs: PM compliance, schedule compliance, planned-work %, MTBF, MTTR.
    - OEE: Availability × Performance × Quality.
7. **ERPNext (v16, confirmed) already provides most of a stock and purchasing backbone.** It has a REST API for every DocType, Material Requests (Purchase/Transfer/Issue), Stock Entries (Material Issue/Receipt/Transfer, custom Stock Entry Types), per-warehouse auto-reorder, Bin stock levels and HMAC-signed webhooks. Its own Asset Maintenance module covers only basic periodic tasks and logs, and Asset Repair records repairs. v16 extends stock reservation to manufacturing Work Orders, Production Plans and subcontracting, but still not to maintenance work, so the CMMS keeps its own demand holds. So the CMMS should drive maintenance and leave stock and valuation in ERPNext (Section 9).

Top recommendations

- Build the MVP around:
    - asset registry and hierarchy
    - QR-based work requests and work orders
    - time-based and meter-based PM
    - standard checklists (SOPs)
    - spares issue and return against work orders, posted to ERPNext
    - downtime and failure codes
    - a KPI dashboard

  Build **multi-site and role-based access control (RBAC) into the first release**, since three sites are confirmed.
- Build the **technician UI as a phone-first, offline-capable Progressive Web App (PWA)** in Vue: bilingual **Tamil/English**, large touch targets, photo capture and QR scanning. Give **engineers rich desktop workspaces** for planning, analytics and configuration, in the same codebase.
- For **spares, use Option A (now confirmed): ERPNext v16 is the stock and valuation master.** The CMMS pulls items, UOMs, groups, warehouses and Bin levels (read-mostly) and shows availability across the master store and sub-stores. Technicians request; storekeepers issue and return against WOs, posted as ERPNext Stock Entries; inter-store moves are Material Transfers. Shortages become Admin-approved Material Requests. Sync uses a transactional outbox, idempotency keys, webhooks and mandatory pull reconciliation (Section 9).

- **Model where each repair happens** (in-situ, Mechanical shop, Electrical shop, external vendor) on the WO, with workshop repair jobs and queues. Start external repairs with a stock transfer to an "At Vendor for Repair" warehouse plus a service PO (Option R1); move to ERPNext Subcontracting (R2) if volume or GST job-work reporting requires it.
- **Self-host on-premise** with the existing PyCharm → Ansible pipeline. Serve the PWA over **HTTPS with a certificate trusted by all phones**: a publicly trusted certificate for an internal hostname via ACME DNS-01 is recommended, because service workers need a secure context. Connect the Ancillary Plant and Campus via site-to-site VPN (Section 8.4).
- Use **ISO 14224-style failure coding** and **EN 13306 terms** for maintenance types from day one. Consistent data is what makes MTBF/MTTR and Pareto analysis credible later.
- Standardise through a **central task/checklist library** (global templates with site-specific variants). Support **TPM autonomous maintenance (CIL: clean/inspect/lubricate) checklists** for operators.
- Defer IoT/condition-based triggers, reorder-level optimisation, repairable-spares cycles and advanced analytics to later phases. Design the data model and API so these can be added cleanly.

2. Competitive analysis

2.1 Products reviewed

Product	Vendor / type	Positioning (per sources)
Maintain X	Commercial SaaS CMMS	Mobile-first CMMS: work orders, PM, procedures, assets, inventory. Free Basic tier; Essential $20 and Premium $65 per user/month billed annually; Enterprise custom [S1–S3]
UpKeep	Commercial SaaS CMMS	Mobile-first CMMS with "Nova AI". Essential $24 and Premium $55 per user/month; Professional and Enterprise quoted; unlimited free requesters and view-only users [S4–S6]
Fiix	Rockwell Automation, SaaS CMMS	Cloud CMMS. Free, Basic $45 and Professional $75 per user/month; Enterprise custom. Integrates with FactoryTalk Optix for condition-based maintenance [S7–S9]
Limble	Commercial SaaS CMMS	Standard, Premium+ and Enterprise plans. Limble's own cost guide lists Standard at $28 and Premium+ at $69 per user/month [S10–S13]
eMaint	Fluke Reliability, SaaS CMMS/EAM	Team $69 and Professional $85 per user/month (3-user minimum); Enterprise quoted. Multi-site and multi-language [S14–S17]
IBM Maximo Application Suite	IBM, EAM/APM	Manage (EAM), Monitor (IoT), Health, Predict. Client-managed or SaaS. Licensed through "AppPoints" credits [S18–S21]
SAP PM / S/4HANA Asset Management	SAP, ERP-integrated EAM	Notifications, maintenance orders, maintenance plans, functional locations/equipment. SAP Service and Asset Manager for mobile [S22–S26]
Infor EAM → **HxGN EAM**	Hexagon (acquired Infor's EAM business, Oct 2021)	Work management (planned, corrective, preventive, inspection), unlimited parent/child hierarchy, offline mobile [S27–S30]
Odoo Maintenance	Module of the Odoo ERP suite (open-source core; edition/licensing to be confirmed)	Corrective and preventive requests, maintenance calendar, teams, equipment linked to work centres, automatic MTBF/MTTR [S31–S33]
openMAINT	Open source (AGPL-3), built on CMDBuild	Property and facility management: asset inventory, preventive/meter-based/corrective maintenance, warehouse, suppliers/contracts, GIS/BIM. REST/SOAP; mobile app and self-service portal are subscription features [S34–S36]

2.2 Feature comparison matrix

Legend: ✓ = supported per cited source (tier noted where the source states it); **n/v** = not verified in the sources reviewed. *n/v does **not** mean the product lacks the feature.*

Capability	MaintainX	UpKeep	Fiix	Limble	eMaint	Maximo	SAP PM	HxGN EAM	Odoo	openMAINT
Work requests (free requesters / portal / QR)	✓ free requesters; request portals with QR/web link	✓ free requesters; external portal (Professional)	✓ unlimited service requests (Free)	✓ portal via QR/URL/email, no login (Standard)	✓ request-only portal licences (unlimited in Enterprise)	n/v	✓ notifications	✓ work requests	✓ maintenance requests	✓ self-service portal (subscription)
Work orders	✓	✓	✓	✓	✓	✓	✓ maintenance orders	✓ incl. child WOs	✓	✓
Time-based PM	✓ repeating WOs (Essential)	✓ PM scheduling (Premium)	✓ 25 active PMs Free; unlimited Basic; nested PMs (Pro)	✓ (Standard)	✓ (Team)	✓	✓ maintenance plans	✓	✓ preventive requests	✓
Meter/usage-based PM	✓ (Premium)	n/v	n/v	✓ meter & threshold (Premium+)	✓ meter readings (Team)	✓ meter data for CBM	✓ measurement readings (mobile)	✓ meter readings	n/v	✓ meter-based
Condition-based / IoT triggers	✓ IoT sensors (Enterprise)	✓ IoT sensors (per FAQ)	✓ via FactoryTalk Optix hardware integration	✓ IoT (Enterprise)	✓ CBM triggers (Professional)	✓ Monitor + CBM	n/v	n/v	n/v	n/v
Predictive / AI	✓ CoPilot (Enterprise)	✓ Nova AI (all plans)	✓ Foresight (Professional)	n/v	✓ AI Suite (Enterprise)	✓ Predict, Health	n/v	n/v	estimated next failure (MTBF-based)	n/v
Checklists / procedures / SOPs	✓ procedures: tasks, inspection checks, signatures, photos, meter readings	✓ custom checklists (Premium); signature capture (Professional)	✓ task lists; pass/fail inspection (Basic)	✓ instructions/checklists	✓ PM checklists; advanced procedures; shared task library	✓ inspections	✓ checklists (mobile)	✓ checklists, inspections	n/v	n/v
Spare parts inventory	✓ (Premium)	✓ with costing (Premium)	✓ (Free); cycle counts (Pro)	✓ (Premium+); cycle counts (Enterprise)	✓ (Team); forecaster & cycle counts (Pro)	✓ incl. AI inventory optimisation	✓ (ERP)	✓ parts	n/v	✓ warehouse
Purchasing / POs	✓ (Premium)	✓ (Enterprise)	✓ purchasing & RFQs (Pro)	✓ vendor & PO mgmt (Premium+)	✓ purchasing & requisitions (Pro)	n/v	✓ (ERP)	n/v	via other Odoo apps (n/v)	✓ suppliers/contracts
Downtime tracking	n/v	✓ reliability & downtime (Enterprise)	✓ (Free)	✓ automatic downtime reporting (Standard)	✓ (Team)	n/v	n/v	n/v	n/v	n/v
Failure / cause codes	✓ root cause analysis (Enterprise, per GetApp)	n/v	✓ failure codes (Enterprise)	n/v	n/v	✓ failure mode analysis (RCM)	✓ catalogs on notification types	n/v	n/v	n/v

Capability	MaintainX	UpKeep	Fiix	Limble	eMaint	Maximo	SAP PM	HxGN EAM	Odoo	openMAINT
Offline mobile	✓ offline mode	✓ (Professional)	n/v	✓ (Premium+)	✓ (Professional)	✓ Maximo Mobile	✓ SAP Service & Asset Manager	✓ Mobile Offline	n/v	mobile app (subscription; offline n/v)
QR / barcode scanning	✓	n/v	n/v	✓	✓ (listed on pricing page)	n/v	n/v	n/v	n/v	n/v
Asset hierarchy	n/v	n/v	✓ drag-and-drop hierarchies, asset grouping	n/v	n/v	✓ asset registry	✓ functional location → equipment	✓ unlimited parent/child, positions, systems	categories + work centres	✓ asset inventory
Multi-site	✓ (Enterprise)	✓ (Enterprise)	✓ (Professional)	✓ multi-location (Enterprise)	✓ global multi-site (Enterprise)	✓ (case studies: multi-site customers)	n/v	n/v	multi-company	n/v
Audit trail / e-signature	n/v	✓ audit trails (Enterprise, per FAQ)	✓ audit trail, e-signatures (Enterprise)	✓ 21 CFR compliance (Enterprise)	✓ audit trail (Team); e-signatures	n/v	n/v	n/v	n/v	n/v
API / ERP integration	✓ REST API (Premium); SAP integration (extra cost)	✓ API & custom (Enterprise)	✓ Integration Hub / custom API (Enterprise add-on)	✓ REST API (Premium+); ERP (Enterprise)	✓ API: ERP, BI, SCADA (Enterprise)	n/v	native ERP	n/v	native ERP	✓ REST/SOAP
Built-in reliability KPIs (MTBF/MTTR)	✓ advanced analytics	✓ analytics	✓ reports/analytics	✓ dashboards/KPIs	✓ reports & dashboards	✓ Health KPIs	n/v	n/v	✓ automatic MTBF/MTTR on equipment	n/v
Technician certifications	n/v	n/v	✓ (Basic)	n/v	n/v	n/v	n/v	n/v	n/v	n/v
Calibration	n/v	n/v	n/v	n/v	✓ calibrations, labels	n/v	n/v	n/v	n/v	n/v

2.3 Pricing tier insight (the case for building in-house)

Vendor	Entry paid tier	Where the "standardisation/visibility" features first appear
MaintainX	Essential $20/user/mo (annual)	Premium $65: meter PM, inventory, POs, API. Enterprise: multi-site, SSO, IoT, custom permissions [S1–S3]
UpKeep	Essential $24/user/mo	Premium $55: PM, checklists. Professional (quote): offline. Enterprise (quote): multi-site, downtime, POs, API, custom roles [S4–S6]
Fiix	Basic $45/user/mo	Professional $75: multi-site, purchasing. Enterprise: failure codes, audit trail, custom workflows [S7]
Limble	Standard ($28 per Limble guide)	Premium+ ($69): offline, inventory, meter PM, API. Enterprise: multi-location, custom roles, ERP/IoT [S10, S12, S13]
eMaint	Team $69/user/mo, 3-user minimum	Professional $85: offline, CBM, workflows, purchasing. Enterprise: multi-site, SSO, API [S14, S15, S17]

With ~500 machines over three sites, the features this project needs most are multi-site, failure codes, audit trail, inventory, offline mobile and custom roles. Every vendor reviewed puts these in its top or second-highest tier. That supports building in-house, as long as the team can carry the long-term cost of maintaining the software itself **[ASSUMPTION — validate: internal development/support capacity]**.

2.4 Design and UX takeaways

1. **One-tap requests from the shop floor.** A QR code on each machine opens a pre-filled request form with the asset already linked, as in Limble [S11] and MaintainX [S2]. Use no login, or a lightweight operator login.
2. **Requester seats are free** in MaintainX, UpKeep and Limble [S1, S5, S10]. In-house there is no licence limit, so give *every* operator request access.
3. **Treat procedures as first-class objects.** They attach to WOs and PMs and have step types such as checkbox, pass/fail, number/meter reading, photo and signature. See MaintainX procedures [S2], Fiix pass/fail inspections [S7] and the eMaint shared task library [S15].
4. **Make the technician app offline-first,** with cached assigned work and automatic sync. See MaintainX [S2], Limble [S11], SAP Service and Asset Manager [S26], Maximo Mobile [S21] and HxGN EAM Mobile [S29].
5. **Keep the "problem" separate from the "work":** SAP's notification vs. order [S22, S23] and HxGN's request vs. WO [S27]. This keeps request triage clean and keeps failure coding on the problem record.
6. **Group similar assets and schedule them once.** Fiix lets teams group similar assets so tasks are scheduled, assigned and tracked once [S8]. This suits standardisation across ~500 machines with many repeated types.
7. **Show reliability stats on the asset page.** Odoo shows MTBF, MTTR, latest failure and estimated next failure on each equipment record [S31].
8. **Due-date reminders, push/email and a daily digest are the baseline.** Automatic escalation is an advanced feature (Enterprise in UpKeep [S5]). In-house, include basic escalation early, because it drives response discipline.
9. **Pilot, then roll out.** eMaint suggests starting with a small pilot, then expanding to other teams and locations [S16]. MaintainX says implementing one site takes its average customer about three weeks [S1]. In-house: pilot at the Main Plant, then roll out to the other two sites.
10. **Use multiple languages to standardise across sites.** eMaint highlights multi-site and multi-language capability for standardising work [S16]. In-house: Tamil/English from the start.

2.5 Standards and best-practice inputs

Source	What we take from it
ISO 55000:2024: Asset management, vocabulary, overview, principles [S37, S38]	Principles of value, alignment and leadership; asset decisions aligned to organisational objectives. Positions the CMMS as the data backbone of an asset-management system (ISO 55001 holds the requirements).
ISO 14224:2016: Reliability and maintenance data for equipment [S39, S40]	Equipment taxonomy: "equipment unit" is level 6, and "item" covers levels 6–9 (the equipment unit and its subdivisions). Definitions: **failure mode** = manner in which failure occurs; **failure mechanism** = process that leads to failure (physical, chemical, logical, or a combination); **detection method** = method or activity by which a failure is discovered; **failure impact** classes at equipment level = critical / degraded / incipient. *ISO 14224's formal scope is the petroleum, petrochemical and natural gas industries, so we adopt its data structure, not its equipment-class code lists.*
EN 13306:2017: Maintenance terminology [S41, S42]	Maintenance types: preventive (at predetermined intervals or to prescribed criteria, to reduce probability of failure/degradation), corrective (to restore an item after a failure), condition-based (preventive maintenance that includes assessment of physical condition), predictive (condition-based

Source	What we take from it
	maintenance following a forecast). Used as the WO type list.
SMRP Best Practices metrics [S43, S44]	Standard KPI definitions: PM/PdM compliance, schedule compliance (preferably labour-hours based), planned-work %, MTBF, MTTR, OEE.
OEE [S45]	OEE = Availability × Performance × Quality. Availability = Run Time / Planned Production Time; Performance = (Ideal Cycle Time × Total Count) / Run Time; Quality = Good Count / Total Count. Also the Six Big Losses.
TPM (JIPM) [S46, S47]	Eight pillars, including Autonomous Maintenance and Planned Maintenance. Autonomous maintenance is built on CIL (cleaning, inspection, lubrication) standards, visual management and one-point lessons (OPLs); 5S is the foundation.
India DPDP Act 2023 / DPDP Rules 2025 [S48, S49]	Employee personal data (names, phone numbers, labour time logs, photos) falls under DPDP. Notice, security safeguards and breach notification obligations apply.

3. Scope, stakeholders and user roles

3.1 Scope

In scope (product):

- asset registry and hierarchy for three sites
- work requests and work orders
- PM (time, meter and condition based)
- standard checklists/SOPs, including TPM autonomous-maintenance checks
- maintenance spares management integrated with **ERPNext**, which stays the stock and valuation master (Section 9)
- procurement hooks into ERPNext (Admin-approved Material Requests; PO/receipt status tracking)
- repair locations and workshops (in-situ, Mechanical shop, Electrical shop, external vendor) with repair jobs
- downtime and failure tracking
- meter readings
- scheduling/calendar
- mobile PWA with QR
- notifications and documents
- dashboards and KPIs
- audit trail
- multi-site
- users/RBAC
- Tamil/English UI

Out of scope (initially):

- purchasing, stock ledger/valuation and stock counts, which stay in **ERPNext**
- accounting, which stays in **Tally**; **Tally integration is out of scope** for now
- ERPNext Asset Maintenance module: not used for plant maintenance (see §9.1)
- legacy data migration: **greenfield**, nothing to migrate (bulk import templates are still provided for initial onboarding)
- payroll/HR
- production planning/MES
- fixed-asset depreciation
- predictive ML models (we capture the data now and build models later)

3.2 Sites and scale (confirmed)

Site	Code (proposed)	Role	Notes

Site	Code (proposed)	Role	Notes
Main Plant	MP	Primary manufacturing site	Expected to hold the largest share of the ~500 machines (split **[ASSUMPTION — validate]**)
Ancillary Plant	AP	Supporting manufacturing	—
Campus	CP	Smaller site	What it hosts (utilities/facilities and/or smaller production) is **[ASSUMPTION — validate]**

Total: about 500 machines/equipment (confirmed). Sizing in Section 5 is based on this figure.

3.3 Stakeholders

Stakeholder	Interest
Owner / management	Visibility of downtime, cost and compliance across all three sites; standardisation
Plant managers (per site)	Availability, OEE, maintenance backlog, cost
Maintenance function (supervisors, planners, technicians)	Daily execution, planning, history
Production (operators, shift in-charges)	Fast breakdown reporting, machine availability, autonomous-maintenance checks
Stores / purchase (ERPNext users)	Spares availability, consumption, reorder, Material Requests and POs in ERPNext
IT / software team	Maintainability, security, on-prem hosting, Ansible deployment, backups, ERPNext integration
Safety / quality	LOTO/permit adherence, audit trail **[ASSUMPTION — validate whether a safety/quality function will use the system]**

3.4 User roles

Role	Primary device	Key permissions (summary)
Operator (requester)	Phone/shared tablet, QR	Raise work requests, see status of own requests, complete assigned autonomous-maintenance/CIL checklists
Technician	**Phone (primary)**: phone-first PWA, works offline	View and carry out assigned WOs; record labour, readings, photos, failure codes; **request** parts and record used qty (**cannot issue stock**); raise follow-up WOs
Workshop technician (Mechanical shop / Electrical shop)	Phone + shared workshop desktop	Work the workshop repair-job queue; record labour, parts requested, findings; mark repaired or recommend scrap
Maintenance engineer / supervisor	**Mobile + desktop**: mobile on the floor, rich desktop views for planning/analytics	Triage requests, create/assign WOs, approve completion, manage team workload, handle escalations
Planner (engineer)	**Desktop (primary) + mobile**	Maintain PM plans, task/checklist library, schedule calendar, parts kitting, backlog
Storekeeper (per storeroom: master store or a sub-store)	Desktop + scanner/phone	**The only role that issues parts**; confirms returns; inter-store transfers (dispatch/receive); kitting; repairable receipts and dispatch to workshops/vendors; ERP exceptions queue. Stock ledger, receipts and counts stay in ERPNext
Plant manager	Desktop/mobile dashboards	Site dashboards and KPIs, read access to all site data (no spares/purchase approvals; these are the Admin's)
Corporate viewer / management [ASSUMPTION]	Desktop	Cross-site read-only dashboards
Administrator (Admin), exactly **one per software instance**, covering all sites	Desktop + mobile approval inbox	Users, roles, sites, master data, code lists, configuration, audit log access. **Single approver** for maintenance Material Requests (Purchase), external repairs, reorder-level pushes and scrap in the CMMS, and for maintenance POs in ERPNext; may optionally name a leave-cover delegate (FR-APR-003). No per-site admins and no second approver

A user can hold different roles at different sites (for example, supervisor at the Ancillary Plant and viewer at the Main Plant).

Devices (confirmed): technicians do most of their work on **phones**, so the mobile PWA is their primary UX and every technician flow must be fully usable on a phone. **Engineers** (supervisors, planners, reliability) use **both mobile and desktop**. Desktop provides rich multi-pane views for planning, analytics and configuration (FR-DSK-xxx).

4. Functional requirements

Priority uses **MoSCoW**: M = Must, S = Should, C = Could, W = Won't (for now). **MVP** = included in Release 1.

4.1 Asset registry and hierarchy (AST)

The hierarchy has configurable level names: **Organisation → Site (MP / AP / CP) → Area/Department → Line/System → Machine (equipment unit) → Component/Sub-assembly → Maintainable item**.

- The "Machine" level matches ISO 14224's equipment unit (taxonomy level 6).
- Components and maintainable items match the lower ISO 14224 "item" levels.
- At confirmed scale, the tree holds 3 sites and about 500 machine-level nodes, plus their components.
- Example areas for a tile plant (body preparation, pressing, drying, glazing, kiln/firing, sorting/packing, utilities) are **[ASSUMPTION — validate against the actual plant layout]**.

ID	Requirement	Pri	MVP
FR-AST-001	The system shall store assets in a tree of unlimited depth. Each node is typed: Site, Area, Line, Machine, Component or Item.	M	✓
FR-AST-002	Each asset shall have an asset code that is unique across all sites and follows a configurable pattern (e.g., `MP-PRS-L1-PRESS02`).	M	✓
FR-AST-003	Asset attributes shall include: name (EN/TA), class/category, make, model, serial no., year, commissioning date, supplier, warranty expiry, criticality (A/B/C), status (in service, standby, under repair, decommissioned), cost centre, location.	M	✓
FR-AST-004	Admins shall define asset classes (e.g., hydraulic press, ball mill, kiln, compressor, conveyor) with class-specific custom fields and a default failure-code set.	M	✓
FR-AST-005	Users shall be able to move assets within the hierarchy and across sites (drag-and-drop or form). Move history is kept.	S	
FR-AST-006	The asset page shall show WO history, PM plans, open requests, meter readings, downtime, spare parts (BOM), documents and MTBF/MTTR.	M	✓
FR-AST-007	Each asset shall support a spare parts list / bill of materials linked to inventory parts.	S	✓
FR-AST-008	Bulk import/export of assets through Excel/CSV templates, with a validation report. This is the main onboarding path for a **greenfield** start: the initial load of ~500 machines and their components across the 3 sites, with no legacy data to migrate.	M	✓
FR-AST-009	The system shall generate printable QR labels (single and batch, by site/area/line), each encoding a URL to the asset.	M	✓
FR-AST-010	Criticality assessment: a simple scoring matrix (safety, production impact, quality, cost) shall derive criticality A/B/C.	S	
FR-AST-011	Asset grouping: tag similar assets (same class/model, including across sites) so PM plans and checklists can be applied to a whole group at once.	S	✓
FR-AST-012	Decommissioned assets keep their full history and are hidden from default lists.	M	✓
FR-AST-013	Track rotable/repairable components (e.g., motors, gearboxes) that move between machines, with install/remove history (linked to ERPNext serial numbers; see FR-SPR-017).	S	
FR-AST-014	Downloadable onboarding template pack: assets/hierarchy, asset classes and custom fields, meters, PM plans, checklist templates, spares/BOM links (see FR-SPR-020). Each has sample rows, data validation (drop-downs) and bilingual columns. Re-importable for corrections (upsert by code).	M	✓

4.2 Work requests (REQ)

ID	Requirement	Pri	MVP
FR-REQ-001	Any authenticated operator shall be able to raise a request by scanning an asset's QR code or picking the asset from the hierarchy.	M	✓
FR-REQ-002	A request shall capture: asset, problem description, optional symptom code, machine state (running / degraded / stopped), photos, requester, timestamp.	M	✓
FR-REQ-003	If machine state = stopped, the system shall automatically open a downtime event starting at the	M	✓

ID	Requirement	Pri	MVP
	request time.		
FR-REQ-004	Requests shall be routed to the supervisor queue for the asset's site/area.	M	✓
FR-REQ-005	Supervisors shall approve (convert to WO), merge as duplicate, or reject with a reason. The requester is notified.	M	✓
FR-REQ-006	Duplicate detection: warn when an open request/WO already exists for the same asset.	S	✓
FR-REQ-007	Requesters shall see the status of their requests (submitted, accepted, in progress, done, rejected).	M	✓
FR-REQ-008	Optional kiosk/shared-device mode with PIN login for operators without personal phones.	S	
FR-REQ-009	Requests for non-asset issues (building, safety) raised against a location node.	S	

4.3 Work orders (WO)

ID	Requirement	Pri	MVP
FR-WO-001	WO types aligned with EN 13306 (configurable list): Corrective (immediate/breakdown), Corrective (deferred), Preventive (time), Preventive (meter), Condition-based, Inspection, Autonomous maintenance (TPM), Improvement/Modification, Safety.	M	✓
FR-WO-002	WO fields: number, type, site, asset, priority (P1–P4), description, requested by, assigned to (person/team), planned start/due, estimated hours, status, checklist(s), required parts, **repair location**, safety notes, attachments.	M	✓
FR-WO-003	Status lifecycle: Draft → Approved → Assigned → In Progress → On Hold (reason: parts / production / permit / other) → Completed → Verified/Closed, plus Cancelled. Each transition is role-controlled.	M	✓
FR-WO-004	Technicians shall record labour time (start/stop timer or manual entry). A WO can have multiple technicians.	M	✓
FR-WO-005	Technicians shall request parts and record used quantities on the WO; storekeepers issue and confirm returns, which post to ERPNext and reduce stock there (FR-SPR-007…010).	M	✓
FR-WO-006	Closing a corrective WO shall require failure data: failure mode, cause and action taken (ISO 14224-style code lists), plus free-text remarks.	M	✓
FR-WO-007	Supervisors shall verify completion before closure (optional, configured by WO type/priority).	S	✓
FR-WO-008	A follow-up WO can be created from an existing WO and is linked as parent/child.	S	✓
FR-WO-009	WO cost roll-up: labour hours × rate + parts + external service cost.	S	
FR-WO-010	External/contractor work: record vendor, service report and cost; external component repairs use repair jobs (FR-WSH-004, FR-SPR-023).	S	
FR-WO-011	Safety: for configured asset classes, a LOTO/permit-to-work checklist step must be completed before the WO can move to "In Progress".	S	
FR-WO-012	Multi-asset WO (e.g., one inspection round covering several machines).	C	
FR-WO-013	Printable bilingual WO job card (A4/A5) as a paper fallback.	S	✓
FR-WO-014	Each WO shall record `repair_location` = In-situ | Mechanical shop | Electrical shop | External (configurable list), defaulting to In-situ, plus workshop or vendor where relevant.	M	✓
FR-WO-015	A component removed from a machine can be sent for repair as a **repair job** (child of the WO) to a workshop or vendor, with its own status, labour, parts and cost, rolled up to the parent WO and the asset/serial history.	S	

4.4 Preventive maintenance (PM)

ID	Requirement	Pri	MVP
FR-PM-001	A PM plan shall link an asset (or asset group) to a task list/checklist, WO type, trade/team, estimated duration and required parts.	M	✓
FR-PM-002	Time-based triggers: every N days/weeks/months, specific weekdays, or calendar dates. Schedule can be fixed (from due date) or floating (from last completion).	M	✓
FR-PM-003	Meter-based triggers: every N units (hours, cycles, strokes, tonnes, m^2), based on meter readings.	M	✓
FR-PM-004	Combined trigger: whichever comes first, time or meter.	S	✓
FR-PM-005	Condition-based triggers: create a WO when a reading crosses a threshold (manual readings in MVP; sensors/IoT later).	S	
FR-PM-006	Lead time: create PM WOs X days before the due date. Show a forecast of upcoming PMs (30/60/90 days).	M	✓
FR-PM-007	Nested/cascading PMs (e.g., the monthly task is suppressed when the quarterly task is due).	C	

ID	Requirement	Pri	MVP
FR-PM-008	Configurable PM compliance window per plan (e.g., ±10% of interval). Each PM is recorded as on-time, late or missed.	M	✓
FR-PM-009	Seasonal/shutdown PMs linked to planned shutdown windows.	C	
FR-PM-010	Apply a PM plan template to all assets in a group/class, across all three sites, in one action.	S	✓

4.5 Checklists, SOPs and standardisation (CHK)

ID	Requirement	Pri	MVP
FR-CHK-001	A central library of checklist templates with versioning (draft / published / retired).	M	✓
FR-CHK-002	Step types: checkbox, pass/fail/NA, numeric with min/max limits, text, photo required, multiple choice, signature, and meter reading (which updates the meter).	M	✓
FR-CHK-003	A failed step or out-of-limit reading can optionally create a follow-up request/WO automatically.	S	✓
FR-CHK-004	Templates are scoped as Global (all sites) or Site-specific. A site variant derived from a global template keeps its link to that template.	S	✓
FR-CHK-005	Steps shall support bilingual text (EN/TA) and reference images/OPLs (one-point lessons).	M	✓
FR-CHK-006	TPM autonomous maintenance: CIL (clean, inspect, lubricate) checklists assignable to operators per shift/day, with abnormality tagging.	S	✓
FR-CHK-007	5S audit checklists per area, with scoring and trends.	C	
FR-CHK-008	SOP documents (PDF/video) linked to asset classes and checklist steps.	S	
FR-CHK-009	Change approval: publishing a new template version needs planner/supervisor approval. WOs already in progress keep the version they started with.	S	

4.6 Spare parts and inventory (INV): superseded

FR-INV-001…010 from v0.1 are **replaced by FR-SPR-001…024 in Section 9.8**. ERPNext is the stock and valuation master; the CMMS manages maintenance demand and consumption.

4.7 Procurement hooks (PRC): superseded

FR-PRC-001…005 from v0.1 are **replaced by FR-SPR-012…014, FR-APR-001…004 and FR-ERP-001…012 in Section 9.8**. Purchase Orders and receipts remain in ERPNext; maintenance purchase demand is approved by the Admin.

4.8 Downtime and failure tracking (DFT)

ID	Requirement	Pri	MVP
FR-DFT-001	Downtime events per asset: start, end, duration, type (breakdown, planned maintenance, setup/changeover, no material, no operator, power failure, other), linked WO.	M	✓
FR-DFT-002	Downtime starts automatically from “machine stopped” requests and ends at the WO’s “machine restored” step. Manual corrections are audited.	M	✓
FR-DFT-003	Failure code hierarchy per asset class, structured on ISO 14224 concepts: • **failure mode** (e.g., fails to start, overheating, abnormal vibration, leakage) • **failure mechanism** (e.g., wear, fatigue, corrosion, electrical short) • **failure cause** (design, installation, operation, maintenance, material) • **detection method** (operator observation, PM inspection, condition monitoring) • **failure impact** (critical / degraded / incipient) • **maintenance action** (replace, repair, adjust, clean, lubricate)	M	✓
FR-DFT-004	Record the failed component (a lower hierarchy level) against the machine-level WO.	S	✓
FR-DFT-005	Pareto reports: top assets/failure modes by count and by downtime hours.	M	✓
FR-DFT-006	Root cause analysis record (5-Why / fishbone) linked to repeated or critical failures.	C	
FR-DFT-007	Repeat-failure flag when the same asset/failure mode recurs within N days.	S	

4.9 Meter readings (MTR)

ID	Requirement	Pri	MVP
FR-MTR-001	Assets shall have one or more meters (running hours, cycles, strokes, output, temperature, vibration, etc.), each with a unit and type (cumulative vs. gauge).	M	✓
FR-MTR-002	Readings are entered manually (web, mobile or checklist step) with validation: a cumulative meter cannot decrease unless a meter reset is recorded.	M	✓

ID	Requirement	Pri	MVP
FR-MTR-003	Readings can be imported through CSV/API (later from PLC/SCADA/historian).	S	
FR-MTR-004	Trend chart with limit lines per meter.	S	✓
FR-MTR-005	Meter-reading rounds: a scheduled route of readings across assets.	C	

4.10 Scheduling and calendar (SCH)

ID	Requirement	Pri	MVP
FR-SCH-001	Calendar view (day/week/month) of WOs and PMs, filterable by site, area, team and technician.	M	✓
FR-SCH-002	Drag-and-drop rescheduling and reassignment by planner/supervisor.	S	✓
FR-SCH-003	Shift calendar per site and technician availability (leave, shift).	S	
FR-SCH-004	Capacity view: planned vs. available hours per team per week.	S	
FR-SCH-005	Weekly schedule freeze, with schedule compliance measured against the frozen schedule.	C	
FR-SCH-006	Backlog view with age and estimated hours (weeks of backlog).	S	✓
FR-SCH-007	Production downtime windows (planned stops) to schedule PMs into.	C	

4.11 Mobile and QR (MOB)

ID	Requirement	Pri	MVP
FR-MOB-001	Responsive Vue PWA installable on Android phones/tablets, with no app-store dependency. This is the **primary interface for technicians**.	M	✓
FR-MOB-008	Phone-first technician shell: bottom navigation, one-handed use, ≤ 3 taps from home to start an assigned WO, minimum supported width 360 px. Every technician flow (WO execution, checklists, parts request/return, readings, requests) is completable on a phone.	M	✓
FR-MOB-002	Scanning a QR code with the device camera opens the asset page with quick actions: Raise request, Start WO, Record reading, View history, View documents.	M	✓
FR-MOB-003	Offline mode: assigned WOs, checklists and asset data for the user's site(s) are cached. Actions are queued and synced when back online.	M	✓
FR-MOB-004	Photo capture with client-side compression, attachable to a request, WO or checklist step.	M	✓
FR-MOB-005	"My work" home screen: today's WOs by priority, overdue and in-progress.	M	✓
FR-MOB-006	Voice notes / speech-to-text in Tamil and English.	C	
FR-MOB-007	Glove-friendly UI mode with large touch targets; high-contrast theme.	S	✓

4.12 Notifications (NTF)

ID	Requirement	Pri	MVP
FR-NTF-001	In-app notifications for assignment, status change, approval needed and comments/mentions.	M	✓
FR-NTF-002	Web push (PWA) and email notifications, configurable per user.	S	✓
FR-NTF-003	WhatsApp/SMS notifications for P1 breakdowns through a business messaging provider.	S	
FR-NTF-004	Escalation rules: if a P1 request is not acknowledged within N minutes, or a WO is overdue by N hours, notify the next level (per site contacts).	S	✓
FR-NTF-005	Daily/shift digest for supervisors and plant managers.	S	
FR-NTF-006	Low-stock and PM-due-soon alerts.	M	✓

4.13 Documents and attachments (DOC)

ID	Requirement	Pri	MVP
FR-DOC-001	Attach files (PDF, images, video, DWG) to assets, asset classes, WOs, checklists and parts.	M	✓
FR-DOC-002	Document library with categories (manual, drawing, SOP, certificate, warranty) and versioning.	S	
FR-DOC-003	Mark documents as available offline.	C	
FR-DOC-004	File size/type limits and virus scanning on upload.	S	✓

4.14 Dashboards and KPIs (KPI)

ID	Requirement	Pri	MVP
FR-KPI-001	**PM compliance %** = PM WOs completed within compliance window / PM WOs due in period × 100 (SMRP-style).	M	✓
FR-KPI-002	**MTBF** = operating time / number of failures, per asset/class/line/site and period.	M	✓
FR-KPI-003	**MTTR** = total repair time / number of repairs.	M	✓
FR-KPI-004	**Downtime hours** and **availability %** per asset/line/site.	M	✓
FR-KPI-005	**Planned-work %** = planned maintenance labour hours / total maintenance labour hours.	S	✓
FR-KPI-006	**Schedule compliance** (hours-based) against the weekly schedule.	C	
FR-KPI-007	**Backlog** (count, hours, age) and **WO aging**.	M	✓
FR-KPI-008	**Reactive vs. proactive ratio** (corrective hours vs. PM/CBM hours).	S	✓
FR-KPI-009	**Maintenance cost** per asset/site/month (labour + parts + services).	S	
FR-KPI-010	**OEE** = Availability × Performance × Quality, when production counts and cycle times are supplied (manual entry or import).	C	
FR-KPI-011	Role-specific dashboards: technician (my work), supervisor (queue, overdue), planner (PM forecast, backlog), plant manager (site KPIs), management (MP/AP/CP comparison).	M	✓
FR-KPI-012	Drill down from any KPI to the underlying WOs/events; export to Excel/PDF.	M	✓
FR-KPI-013	Request response time (request → acknowledged → WO start).	S	✓
FR-KPI-014	Read-only database view or API for BI tools.	C	

4.15 Audit trail (AUD)

ID	Requirement	Pri	MVP
FR-AUD-001	Record every create/update/delete on business entities: who, when, old → new values, and source (web/mobile/API/import).	M	✓
FR-AUD-002	Status transitions and approvals cannot be changed. Corrections are added as new entries, not edits.	M	✓
FR-AUD-003	Log logins/logouts, failed logins and permission changes.	M	✓
FR-AUD-004	Admin-only audit log viewer with filters and export.	S	✓
FR-AUD-005	E-signature (re-authentication) on checklist sign-off for configured templates.	C	

4.16 Multi-plant / multi-site (SIT)

ID	Requirement	Pri	MVP
FR-SIT-001	The system shall support the three confirmed sites (Main Plant, Ancillary Plant, Campus) as top-level nodes. More sites can be added without code changes.	M	✓
FR-SIT-002	Every transactional record (asset, request, WO, stock, downtime, reading) carries a `site_id`. Users see only the sites they are granted.	M	✓
FR-SIT-003	Global masters (asset classes, failure codes, checklist templates, parts master, units) are shared across sites, with site-level overrides where configured.	M	✓
FR-SIT-004	Per-site configuration: working calendar/shifts, WO numbering prefix (e.g., `MP-WO-2026-00001`, `AP-…`, `CP-…`), escalation contacts, storerooms.	M	✓
FR-SIT-005	Cross-site dashboards comparing KPIs for MP, AP and CP.	M	✓
FR-SIT-006	Shared technicians: a technician can be assigned work at more than one site.	S	✓
FR-SIT-007	Inter-site part transfers and asset moves.	S	

4.17 Users and RBAC (USR)

ID	Requirement	Pri	MVP
FR-USR-001	User accounts with name, employee code, mobile, email (optional), preferred language, sites, roles and trades/skills.	M	✓
FR-USR-002	Role-based access control with permissions per module/action, granted per site (a role assignment = user × role × site).	M	✓
FR-USR-003	Pre-defined roles as in Section 3.4. Admins can clone and customise roles.	M	✓

ID	Requirement	Pri	MVP
FR-USR-004	Authentication: username/mobile + password, optional OTP, PIN for kiosk mode.	M	✓
FR-USR-005	Technician skills/certifications (e.g., electrical licence) with expiry dates. Warn on assignment if one is missing or expired.	C	
FR-USR-006	Teams/crews per site and trade (mechanical, electrical, utilities).	M	✓
FR-USR-007	SSO with Google Workspace/Microsoft 365, if used.	C	
FR-USR-008	Users are deactivated, not deleted, so their history is kept.	M	✓

4.18 Integration and configuration (INT)

ID	Requirement	Pri	MVP
FR-INT-001	Versioned REST API (JSON, documented in OpenAPI) for all modules.	M	✓
FR-INT-002	Webhooks/events (WO created/closed, stock low) for downstream systems.	C	
FR-INT-003	IoT/PLC ingestion endpoint for meter/condition readings (MQTT/OPC UA gateway → REST).	C	
FR-INT-004	ERPNext v16 integration: pull of items, UOMs, item groups, warehouses and stock levels; push of Stock Entries and Material Requests. See Section 9 (FR-SPR, FR-ERP). Tally: out of scope.	M	✓
FR-INT-005	Configurable code lists (priorities, hold reasons, downtime types, failure codes) that need no code change.	M	✓

4.19 Desktop workspaces for engineers (DSK)

ID	Requirement	Pri	MVP
FR-DSK-001	Rich desktop workspace (≥ 1366 px) with multi-pane layouts (tree/list + detail + side panel), keyboard shortcuts, bulk edit and inline editing for WOs, assets and PM plans.	M	✓
FR-DSK-002	Planning board: drag WOs onto a technician × day grid with capacity bars, filters by site/area/trade, and backlog side panel.	S	✓
FR-DSK-003	Analytics workspace: configurable KPI dashboards, cross-filtering, drill-down, export to Excel/PDF (extends FR-KPI-011/012).	M	✓
FR-DSK-004	Configuration consoles: asset classes and custom fields, code lists, checklist builder with live **phone preview**, PM plan editor with forecast preview, ERPNext mapping/health.	M	✓
FR-DSK-005	Saved views and filters per user, shareable with a team.	S	✓
FR-DSK-006	Desktop ↔ phone hand-off: open a WO or asset on the phone from the desktop via QR/deep link.	C	

4.20 Repair locations and workshops (WSH)

Confirmed repair locations: **In-situ**, **Mechanical shop**, **Electrical shop**, **External (vendor/subcontract)**. ERPNext movements for each are in §9.4–9.5.

ID	Requirement	Pri	MVP
FR-WSH-001	Workshop master: code, name (EN/TA), type (mechanical / electrical / other), site, team, ERPNext WIP warehouse (for stocked repairables).	M	✓
FR-WSH-002	Repair location on every WO (FR-WO-014) and on every repair job; filters and KPIs by repair location.	M	✓
FR-WSH-003	Workshop queue per workshop (phone and desktop): incoming repair jobs, priority, due date, linked machine WO, unit/serial, status (Received → In repair → Repaired / Scrap recommended / Sent out).	S	
FR-WSH-004	External repair job: vendor (from ERPNext suppliers), estimate, Admin approval, dispatch and expected return date, overdue alerts, vendor report and final cost (§9.5, FR-SPR-023).	S	
FR-WSH-005	Workshop KPIs: turnaround time, backlog, repeat repairs per unit/serial, in-house vs external cost.	C	
FR-WSH-006	Parts used in a workshop are requested against the repair job and issued by the storekeeper like WO parts.	S	

5. Non-functional requirements

5.1 Sizing basis

Quantity	Value	Basis
Sites	3 (Main Plant, Ancillary Plant, Campus)	**Confirmed**
Machines / equipment	~500	**Confirmed**
Asset records incl. components	~2,500–5,000	**[ASSUMPTION]** 5–10 tracked components per machine
Named users	~80–200 (incl. operators)	**[ASSUMPTION — validate headcount]**
Peak concurrent users	~50	**[ASSUMPTION]**
WOs per year (all types)	~15,000–30,000	**[ASSUMPTION]** ~500 machines × (PM + breakdowns + autonomous-maintenance checks)
Checklist step results per year	~0.5–1.5 million	**[ASSUMPTION]**
Attachment growth	~20–60 GB/year	**[ASSUMPTION]** compressed photos
Maintenance spare items in ERPNext	~1,000–3,000	**[ASSUMPTION]** greenfield spares list to be built
Spares transactions (issue/return/transfer) per year	~10,000–25,000 Stock Entries	**[ASSUMPTION]**
Design headroom	10× the above (e.g., 5,000 machines / 50,000 assets, 10+ sites, 500 users, 300k WOs/yr)	Design target

This is a modest scale. A single well-tuned MariaDB instance and two application instances are enough, and no sharding is needed. Site separation is logical (a `site_id` on every record), not physical (separate databases per site).

5.2 NFR table

ID	Category	Requirement	Pri	MVP
NFR-PERF-001	Performance	95th-percentile API response < 500 ms for list/detail calls at 50 concurrent users; dashboards < 3 s with 3-site data.	M	✓
NFR-PERF-002	Performance	First meaningful screen on mobile < 3 s on a mid-range Android phone over 4G; QR scan to asset page < 2 s (online).	M	✓
NFR-PERF-003	Performance	Nightly PM generation for all plans across the 3 sites (~500 machines) finishes in < 2 minutes.	M	✓
NFR-OFF-001	Offline	While offline for at least one shift (8–12 h), a technician can view assigned WOs (next 7 days), complete checklists, log time/parts, take photos and raise requests. Data syncs automatically on reconnect.	M	✓
NFR-OFF-002	Offline	Conflict handling: the server is the authority for status. Offline edits are replayed in order. Conflicts are shown to the user/supervisor and are never silently lost.	M	✓
NFR-AVL-001	Availability	99.5% during plant operating hours at all three sites; planned maintenance outside shifts.	S	✓
NFR-NET-001	Multi-site network	The server is on-premise. Ancillary Plant and Campus reach it via site-to-site VPN or an existing leased link. Only HTTPS (443) is open, and only to site subnets/VPN, with no public internet exposure. If the link drops, sites keep working in PWA offline mode and sync later.	M	✓
NFR-TLS-001	Certificates	The PWA is served only over HTTPS (service workers need a secure context [S74, S75]), using a certificate that every user device trusts **without manual CA installation**. Recommended: a publicly trusted certificate for an internal hostname via ACME DNS-01. Automatic renewal by Ansible; alert ≥ 14 days before expiry.	M	✓
NFR-SEC-001	Security	HTTPS/TLS 1.2+ everywhere, with HSTS.	M	✓
NFR-SEC-002	Security	Passwords hashed (Argon2/bcrypt); lockout after repeated failures; session/JWT expiry and refresh.	M	✓
NFR-SEC-003	Security	RBAC and site scoping enforced on the server for every endpoint, never only	M	✓

ID	Category	Requirement	Pri	MVP
		in the UI.		
NFR-SEC-004	Security	OWASP Top 10 controls: parameterised queries (SQLAlchemy), CSRF protection where cookies are used, input validation, upload type checks.	M	✓
NFR-SEC-005	Privacy	Handle employee personal data in line with India's DPDP Act 2023 / DPDP Rules 2025: notice, purpose limitation, security safeguards and a breach response procedure.	M	✓
NFR-BAK-001	Backup	On-prem backups by the Ansible `backup` role: `mariadb-backup` nightly full + hourly incremental; binary logs retained ≥ 7 days for point-in-time recovery; attachments synced daily. Encrypted copy to a second on-prem location, preferably at another site (e.g., AP), plus a periodic offline/off-site copy. RPO ≤ 1 h, RTO ≤ 4 h. Alert on any failed backup [S78].	M	✓
NFR-BAK-002	Backup	Quarterly restore test (automated Ansible restore playbook into staging), documented.	S	✓
NFR-I18N-001	i18n	Full UI in English and Tamil (vue-i18n), chosen per user. Bilingual master-data fields (name_en, name_ta) for assets, parts, checklist steps and codes.	M	✓
NFR-I18N-002	i18n	MariaDB `utf8mb4` with a Unicode collation; Tamil-capable fonts (e.g., Noto Sans Tamil) bundled; reports/PDFs render Tamil correctly.	M	✓
NFR-I18N-003	i18n	IST timezone by default, DD-MM-YYYY display, optional Indian number grouping (lakh/crore), INR currency.	M	✓
NFR-I18N-004	i18n	Further languages (e.g., Hindi) can be added without code changes.	S	
NFR-ACC-001	Accessibility	Target WCAG 2.1 AA for the web app. Minimum 44×44 px touch targets. Colour is never the only status indicator; icons always have text labels.	S	✓
NFR-ACC-002	Accessibility	Low-literacy support: icon-led quick actions, photo-first request flow, optional voice notes.	S	
NFR-USA-001	Usability	A request can be raised in ≤ 30 s and ≤ 4 taps after a QR scan.	M	✓
NFR-USA-002	Usability	A new technician is productive after < 1 hour of training.	S	✓
NFR-SCL-001	Scalability	Supports 10× the sizing basis without changing the architecture.	S	
NFR-MNT-001	Maintainability	Automated tests (pytest, Vitest, Playwright end-to-end) with ≥ 70% backend coverage on domain logic; CI pipeline; DB migrations via Alembic.	M	✓
NFR-MNT-002	Maintainability	OpenAPI spec generated from code; API versioned (`/api/v1`).	M	✓
NFR-OBS-001	Observability	Structured logs, error tracking, health endpoints and basic metrics (request latency, job failures, sync queue depth per site).	S	✓
NFR-DEP-001	Deployment	Self-hosted on-premise. Deployed from Git-tagged releases with the existing PyCharm → Ansible pipeline, using the roles in §8.4. Native systemd services (Gunicorn, Celery) behind Nginx. No manual steps on the server; DB migrations run by the playbook; rollback ≤ 10 min via the release symlink.	M	✓
NFR-DEP-002	Deployment	Staging environment (separate Ansible inventory) with an ERPNext test site. Every release goes to staging and passes the automated tests first.	S	✓
NFR-ERP-001	Integration resilience	The CMMS stays fully usable when ERPNext is down: issues/returns/MRs are queued, and stock is shown from cache with an "as of" time. Queued postings flush within 5 min of ERPNext recovery.	M	✓
NFR-ERP-002	Integration consistency	Within 24 h, 100% of CMMS issue/return lines have either a matching submitted ERPNext Stock Entry or an open exception (nightly reconciliation). No duplicate postings (idempotency).	M	✓
NFR-ERP-003	Integration security	Least-privilege ERPNext integration user; API and webhook secrets in Ansible Vault; HMAC verification of all webhooks; integration traffic over HTTPS on the internal network.	M	✓
NFR-UX-001	Devices	Technician UI optimised for Android phones 360–430 px wide (primary). Engineer desktop UI optimised for ≥ 1366×768 and usable on tablets.	M	✓
NFR-RET-001	Data retention	Maintenance history kept ≥ 10 years (asset life); audit logs ≥ 3 years **[ASSUMPTION — validate]**.	S	

6. Use cases

UC-01 Raise a breakdown request via QR

- **Actor:** Operator.

- **Preconditions:** Operator is logged in (or uses a kiosk PIN); the asset has a QR label.
- **Main flow:**
    1. Operator scans the QR code.
    2. System opens the asset page with “Report problem”.
    3. Operator selects machine state (stopped/degraded/running) and symptom, and adds a photo/description.
    4. Operator submits.
    5. System creates the request, opens a downtime event if the machine is stopped, and notifies the area supervisor at that site. A stopped machine with criticality A escalates as P1.
- **Alternates:**
    - 1a. QR damaged: search for the asset by code/name.
    - 4a. Offline: the request is queued and submitted on reconnect, keeping its original timestamp.
    - 5a. An open request already exists: the system shows it and lets the operator add a comment instead.
- **Postconditions:** Request is in the supervisor queue; downtime is running if the machine is stopped.

UC-02 Triage request and create work order

- **Actor:** Maintenance supervisor.
- **Preconditions:** Request is in the queue for the supervisor’s site/area.
- **Main flow:**
    1. Supervisor opens the queue, sorted by priority/age.
    2. Reviews the request.
    3. Converts it to a WO and sets type, priority, assignee and due date.
    4. System notifies the technician and the requester.
- **Alternates:**
    - 3a. Reject with a reason.
    - 3b. Merge into an existing WO.
    - 3c. Convert to a planned WO for the next shutdown.

UC-03 Execute and complete a corrective WO (mobile)

- **Actor:** Technician.
- **Preconditions:** WO is assigned.
- **Main flow:**
    1. Technician opens “My work”.
    2. Starts the WO (timer starts; LOTO step if configured).
    3. Diagnoses the fault and records the failed component, failure mode and cause.
    4. Issues/records parts.
    5. Records the action taken and photos.
    6. Confirms “machine restored” (downtime ends).
    7. Completes the WO.
- **Alternates:**
    - 2a. Offline: all steps are cached and synced later.
    - 4a. Part not in stock at this site: check stock at other sites, put the WO on hold (parts), raise an inter-site transfer or an ERPNext Material Request (Purchase).
    - 6a. Temporary fix: create a follow-up WO.

UC-04 Verify and close a WO

- **Actor:** Supervisor.
- **Main flow:** Supervisor reviews the completed WO, checks failure codes and parts, and approves it, which moves it to Closed.
- **Alternate:** Supervisor returns the WO to the technician with a comment.

UC-05 Define a PM plan from a template for an asset group (cross-site)

- **Actor:** Planner.
- **Preconditions:** Checklist template is published; asset group exists.
- **Main flow:**
    1. Planner creates a PM plan: selects the template and the trigger (e.g., every 250 running hours or 30 days, whichever comes first), lead time, team and parts.
    2. Applies it to an asset group that spans MP and AP.
    3. System creates a plan instance per asset and shows a forecast per site.
- **Alternates:**
    - 2a. A site-specific variant is chosen for one site.

UC-06 Generate and execute PM WOs

- **Actors:** System (scheduler), Technician.
- **Main flow:**
    1. The nightly job checks time/meter triggers for all sites and creates PM WOs with their checklists, numbered by site prefix.
    2. The technician carries out the checklist. A failed step automatically creates a follow-up request.
    3. On completion, compliance is recorded (on-time/late).
- **Alternates:**
    - Window missed: status becomes Missed and the PM appears on the site compliance report.

UC-07 Record meter readings

- **Actor:** Technician/Operator.
- **Main flow:**
    1. Scan the QR code and choose “Record reading”.
    2. Enter the value.
    3. System validates it, updates the meter, and triggers PM if a threshold is reached.
- **Alternates:**
    - A cumulative meter reading lower than the previous one is rejected unless “meter replaced/reset” is selected.

UC-08 Issue spare parts to a WO

- Superseded by **UC-13** (technician requests, storekeeper issues) in Section 9.9. Related: UC-14 shortage, UC-15 kitting, UC-16 returns, UC-17 workshop repair, UC-18 sync exceptions, UC-19 inter-store transfer, UC-20 external repair, UC-21 Admin approval.

UC-09 Perform an autonomous maintenance (CIL) check

- **Actor:** Operator.
- **Preconditions:** A CIL checklist is scheduled for the machine/shift.
- **Main flow:**
    1. Operator opens “My checks”.
    2. Completes the steps (clean, inspect, lubricate) with photos.
    3. Tags any abnormality; the system raises a request for it.
    4. Completion is tracked per shift.
- **Alternates:**
    - A skipped check is shown to the shift in-charge/supervisor.

UC-10 Review site KPIs and drill down

- **Actor:** Plant manager / management.
- **Main flow:**

1. Opens the dashboard and selects site(s) (MP/AP/CP or all) and period.
2. Views PM compliance, MTBF, MTTR, downtime Pareto and backlog.
3. Drills into a top failure and views the WOs behind it.
4. Exports.

UC-11 Onboard assets in bulk

- **Actor:** Admin/Planner.
- **Main flow:**
  1. Download the import template pack (greenfield start; no legacy data to migrate).
  2. Fill in the hierarchy and attributes for each site (~500 machines in total).
  3. Upload and review the validation report.
  4. Fix errors and commit.
  5. Print QR labels in batches per site/line.
- **Alternates:**
  - Duplicate codes: the rejected rows are listed.

UC-12 Manage users and site roles

- **Actor:** Admin.
- **Main flow:**
  1. Create the user.
  2. Assign roles per site.
  3. Set the language.
  4. The user receives credentials/OTP.
- **Alternate:** Deactivate a leaver and reassign their open WOs.

7. User stories and acceptance criteria

Format: *As a … I want … so that …*, followed by **Given / When / Then** criteria. 42 stories in 9 epics.

Epic E1: Asset registry

US-01 As a planner, I want to import the asset list from Excel so that all ~500 machines across the three sites are online quickly.

- **Given** a template with 500 valid rows covering MP, AP and CP, **When** I upload it, **Then** 500 assets are created under the correct site and hierarchy, and a summary per site is shown.
- **Given** a row with a duplicate asset code, **When** I upload, **Then** that row is rejected with a reason, and no rows are committed until I confirm.

US-02 As a technician, I want to see an asset's full history on one page so that I can diagnose faults faster.

- **Given** an asset with past WOs, **When** I open it, **Then** I see WOs (newest first), downtime, readings, documents and MTBF/MTTR.

US-03 As an admin, I want to print QR labels for a line so that operators can scan machines.

- **Given** I select Line L1 at the Main Plant, **When** I print labels, **Then** a PDF is generated with one label per asset (code, name EN/TA, QR).

US-04 As a planner, I want to define asset classes with custom fields so that each machine type captures the right data.

- **Given** the class "Hydraulic press" has the field "Tonnage", **When** I create a press asset, **Then** "Tonnage" is shown and validated.

US-05 As a plant manager, I want assets flagged by criticality so that A-critical failures escalate faster.

- **Given** an asset of criticality A, **When** a "stopped" request is raised, **Then** priority defaults to P1.

Epic E2: Work requests

US-06 As an operator, I want to report a breakdown by scanning the machine's QR code so that I don't need to know asset codes.

- **Given** I scan a valid QR code, **When** I tap "Report problem", **Then** the asset is pre-filled and I can submit in ≤ 4 taps.

US-07 As an operator, I want to attach a photo so that the technician understands the problem.

- **Given** I take a photo, **When** I submit, **Then** the photo is compressed (< 500 KB) and attached.

US-08 As an operator, I want to see the status of my request so that I know when the machine will be fixed.

- **Given** my request was converted to a WO, **When** I open "My requests", **Then** I see the status "In progress" and the assigned technician.

US-09 As a supervisor, I want duplicate warnings so that we don't open two WOs for one breakdown.

- **Given** an open WO on asset X, **When** an operator reports a problem on X, **Then** the operator sees the open WO and can add a comment instead.

US-10 As an operator without network, I want my request saved so that it isn't lost.

- **Given** the device is offline, **When** I submit a request, **Then** it is queued with its original timestamp, **And** it is sent automatically once the device is back online.

Epic E3: Work orders

US-11 As a supervisor, I want to convert a request to a WO in one step so that response is fast.

- **Given** a request, **When** I click "Create WO" and pick an assignee, **Then** a WO is created, linked to the request and numbered with the site prefix (e.g., `AP-WO-…`), **And** the technician is notified.

US-12 As a technician, I want a start/stop timer so that labour time is accurate.

- **Given** a WO assigned to me, **When** I tap Start and later Stop, **Then** the elapsed time is recorded against me.

US-13 As a technician, I want to pick failure mode, cause and action from lists so that data is consistent.

- **Given** a corrective WO on a "Ball mill" class asset, **When** I complete it, **Then** I must choose failure mode, cause and action from that class's code list.

US-14 As a supervisor, I want to put a WO on hold with a reason so that waiting time is visible.

- **Given** a WO in progress, **When** I set "On Hold – Parts", **Then** the hold reason and time are recorded, **And** the WO appears in the "on hold" report.

US-15 As a supervisor, I want to verify completed WOs so that the quality of work and data is assured.

- **Given** a completed P1 WO, **When** I approve it, **Then** its status becomes Closed.
- **When** I reject it, **Then** it returns to the technician with my comment.

US-16 As a technician, I want to create a follow-up WO so that permanent fixes aren't forgotten.

- **Given** a completed temporary repair, **When** I tap "Follow-up", **Then** a new WO linked to the parent is created in Draft.

Epic E4: Preventive maintenance

US-17 As a planner, I want time-based PM plans so that routine tasks are created automatically.

- **Given** a monthly plan with a 5-day lead time, due 15 Nov, **When** the nightly job runs on 10 Nov, **Then** a PM WO due 15 Nov is created.

US-18 As a planner, I want meter-based PM so that servicing follows actual usage.

- **Given** a 500-hour plan last done at 1,000 h, **When** a reading of 1,500 h is entered, **Then** a PM WO is created.

US-19 As a planner, I want "whichever comes first" triggers so that machines with low usage still get serviced.

- **Given** a trigger of 30 days OR 250 h, **When** 30 days pass with only 100 h used, **Then** the WO is created on the time trigger.

US-20 As a planner, I want to apply one plan to all identical machines across sites so that maintenance is standardised.

- **Given** 12 presses in group "Press-Model-X" at MP and AP, **When** I apply the plan, **Then** 12 plan instances are created, each scheduled independently under its own site.

US-21 As a plant manager, I want PM compliance by site so that I can see how disciplined each site is.

- **Given** 100 PMs were due in October at MP and 92 were completed in their window, **When** I open the dashboard, **Then** PM compliance shows 92% for MP.

US-22 As a planner, I want a 90-day PM forecast so that I can plan labour and parts.

- **Given** active plans, **When** I open the forecast, **Then** I see the expected WOs per week and per site, with estimated hours and parts.

Epic E5: Checklists and standardisation

US-23 As a planner, I want to build checklists with different step types so that inspections are structured.

- **Given** a numeric step "Hydraulic oil temp" with a maximum of 60 °C, **When** a technician enters 65, **Then** the step is flagged out-of-limit.

US-24 As a planner, I want out-of-limit steps to raise a follow-up request so that defects are acted on.

- **Given** auto-follow-up is enabled on a step, **When** the step fails, **Then** a request is created for the asset that references the WO.

US-25 As a planner, I want versioned templates so that changes are controlled.

- **Given** template v1 is used by open WOs, **When** I publish v2, **Then** new WOs use v2 **And** open WOs keep v1.

US-26 As an operator, I want checklist steps in Tamil with pictures so that I can follow them easily.

- **Given** my language is Tamil, **When** I open a CIL checklist, **Then** the steps are shown in Tamil with reference images.

US-27 As a supervisor, I want to see missed operator CIL checks per shift so that autonomous maintenance keeps happening.

- **Given** CIL checks are scheduled for Shift A, **When** the shift ends with checks incomplete, **Then** they appear as Missed on the supervisor dashboard.

Epic E6: Spares and inventory (revised for ERPNext; see also Epic E10 in §9.10)

US-28 As a storekeeper, I want to issue parts to a WO by scanning so that consumption is accurate.

- **Given** a WO and a part with cached ERPNext stock 10 at the master store, **When** I issue 2, **Then** a "Maintenance Issue" Stock Entry is posted to ERPNext, **And** after sync the cached stock is 8 **And** the WO cost includes 2 × the ERPNext valuation rate.

US-29 As a storekeeper, I want low-stock alerts so that critical spares don't run out.

- **Given** an ERPNext reorder level of 5 for the master store and cached stock 6, **When** 2 are issued, **Then** I receive a low-stock alert in the CMMS, **And** ERPNext's auto-reorder (if enabled) raises the Material Request.

US-30 As a technician, I want to see whether a part is available in another store so that repairs aren't delayed.

- **Given** a part is out of stock in the AP sub-store and in stock in the master store (per ERPNext Bin cache), **When** I view it from a WO at AP, **Then** the available quantity in every store (master and sub-stores) with as-of time is shown, **And** I can ask the storekeeper for an inter-store transfer.

US-31 As a supervisor, I want to raise a purchase request from a shortage so that purchasing is triggered.

- **Given** a shortage on a WO, **When** I click "Request purchase" **And** the Admin approves it, **Then** a submitted ERPNext Material Request (Purchase) linked to the WO is created, **And** its status (Ordered/Received) appears on the WO.

Epic E7: Downtime and reliability

US-32 As a plant manager, I want downtime to start automatically from "machine stopped" requests so that downtime data is complete.

- **Given** a request with state "stopped" at 10:05, **When** the WO is marked "machine restored" at 11:20, **Then** a 75-minute downtime event is recorded.

US-33 As a supervisor, I want to correct downtime times with a reason so that data stays accurate and auditable.

- **Given** a downtime event, **When** I edit the end time, **Then** the change and the reason are written to the audit trail.

US-34 As a reliability lead, I want a Pareto of failure modes by downtime so that I can target improvements.

- **Given** failure data for the quarter, **When** I open the Pareto, **Then** failure modes are sorted by downtime hours with cumulative %.

US-35 As a planner, I want MTBF and MTTR per asset class so that I can compare machine types.

- **Given** closed corrective WOs and operating hours, **When** I select a class and period, **Then** MTBF and MTTR are calculated using SMRP-style formulas, **And** I can drill down to the source records.

Epic E8: Notifications, scheduling and mobile

US-36 As a supervisor, I want escalation when a P1 is not acknowledged so that breakdowns get attention.

- **Given** a P1 request at CP unacknowledged for 15 minutes, **When** the rule fires, **Then** CP's configured escalation contact (e.g., plant manager) is notified.

US-37 As a planner, I want to drag WOs on a calendar so that rescheduling is quick.

- **Given** a WO on Tuesday, **When** I drag it to Thursday, **Then** the due date updates **And** the assignee is notified.

US-38 As a technician, I want my assigned work available offline so that I can work in areas with poor signal.

- **Given** I opened the app online in the morning, **When** I lose network, **Then** I can open WOs, complete checklists and record time, **And** on reconnect the changes sync without loss.

Epic E9: Administration, multi-site and audit

US-39 As an admin, I want to grant roles per site so that people only see what they need.

- **Given** a user is Supervisor at AP only, **When** they log in, **Then** they see AP data only **And** cannot access MP or CP WOs through the UI or the API.

US-40 As management, I want a 3-site KPI comparison so that I can compare MP, AP and CP.

- **Given** data for all sites, **When** I open the corporate dashboard, **Then** PM compliance, MTBF, MTTR, downtime and backlog are shown side by side for MP, AP and CP.

US-41 As an admin, I want a complete change history for any WO so that disputes can be resolved.

- **Given** a WO edited several times, **When** I open its history, **Then** every change shows who, when, and the old and new values.

US-42 As a user, I want to switch between Tamil and English so that I can use my preferred language.

- **Given** I set Tamil in my profile, **When** I log in on any device, **Then** the UI and bilingual master data appear in Tamil.

8. High-level architecture

8.1 Component diagram

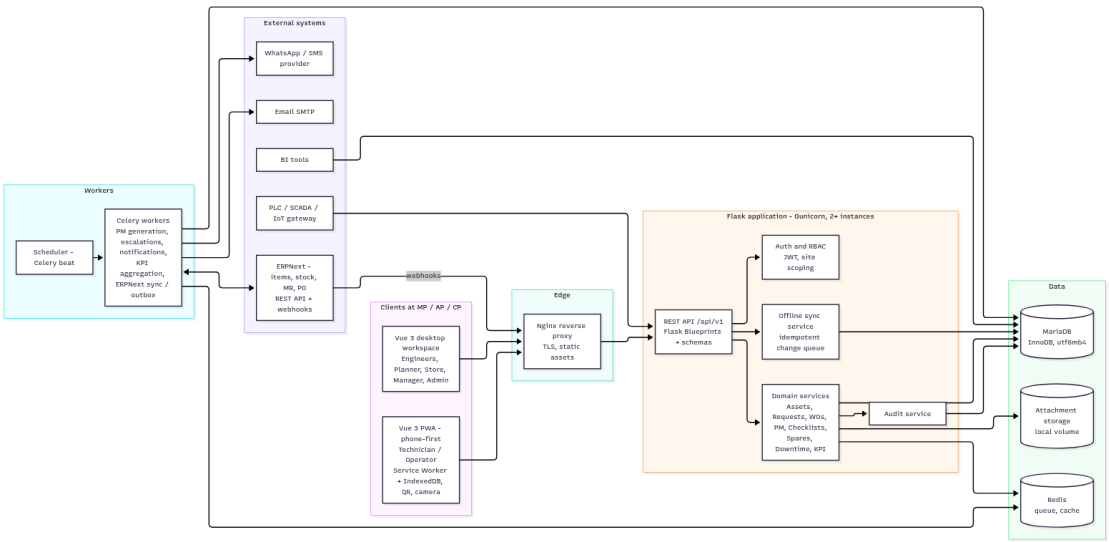


Figure 1 — Component diagram (Vue + Flask + MariaDB + ERPNext)

(Rendered from the mermaid source in the .md version.)

Key design choices

- **Modular monolith** (one Flask Blueprint per module). This fits 3 sites / ~500 machines; split into services later only if needed.
- **One central deployment for all three sites.** Each site is a tenant-like scope (`site_id`), not a separate install. This lets masters, templates and KPIs be standardised across MP, AP and CP.
- **Frontend:** Vue 3 + Vite + Pinia + Vue Router + vue-i18n, with one component library (e.g., Vuetify or PrimeVue).

    - PWA via `vite-plugin-pwa` (Workbox).
    - Offline store in IndexedDB (e.g., Dexie).
    - QR scanning with the camera through a browser library (e.g., `html5-qrcode` / ZXing).
- **Backend:** Flask + SQLAlchemy 2, with Alembic migrations, Marshmallow or Pydantic schema validation, and OpenAPI via `flask-smorest` or `APIFlask`.
- **Background jobs:** Celery or RQ + Redis for PM generation, escalation timers, notifications and nightly KPI aggregation.
- **Offline sync:**
    - The client generates UUIDs for new records and a client mutation ID for each change.
    - `POST /sync/push` applies changes idempotently.
    - `GET /sync/pull?since=<cursor>` returns changes for the user's sites and assignments.
- **Site isolation:** every transactional table has `site_id`. A SQLAlchemy query filter adds the allowed site IDs from the JWT to every query. Automated authorisation tests cover every endpoint for cross-site leakage.
- **Two UX shells, one Vue codebase:** a **phone-first technician shell** (bottom navigation, offline, QR) and a **desktop engineer workspace** (multi-pane planning, analytics, configuration). Both share components, API client, i18n and auth; the route/layout is chosen by role and screen size.
- **ERPNext integration:** a separate `integrations.erpnext` module with transactional outbox, idempotent posting, webhook receiver and reconciliation jobs (Section 9).
- **Hosting:** self-hosted on-premise, deployed by Ansible (Section 8.4).

8.2 REST API module outline (`/api/v1`)

Module	Representative endpoints
Auth	`POST /auth/login`, `POST /auth/refresh`, `POST /auth/otp`, `GET /me`
Sites & org	`GET/POST /sites`, `GET /sites/{id}/calendar`, `GET /sites/{id}/settings`
Assets	`GET/POST /assets`, `GET/PATCH /assets/{id}`, `GET /assets/{id}/tree`, `GET /assets/{id}/history`, `POST /assets/import`, `GET /assets/{id}/qr`, `POST /assets/qr-batch`, `GET/POST /asset-classes`, `GET/POST /asset-groups`
Meters	`GET/POST /assets/{id}/meters`, `POST /meters/{id}/readings`, `GET /meters/{id}/readings`
Requests	`GET/POST /requests`, `POST /requests/{id}/convert`, `POST /requests/{id}/reject`, `POST /requests/{id}/merge`
Work orders	`GET/POST /work-orders`, `GET/PATCH /work-orders/{id}`, `POST /work-orders/{id}/transition`, `POST /work-orders/{id}/labour`, `POST /work-orders/{id}/parts`, `POST /work-orders/{id}/failure`, `GET /work-orders/{id}/history`
PM	`GET/POST /pm-plans`, `POST /pm-plans/{id}/apply`, `GET /pm-plans/forecast`, `POST /pm-plans/run` (admin)
Checklists	`GET/POST /checklist-templates`, `POST /checklist-templates/{id}/publish`, `POST /work-orders/{id}/checklists/{cid}/results`
Spares	`GET /parts` (ERPNext-mirrored + CMMS attributes), `PATCH /parts/{id}` (CMMS attributes only), `GET /parts/{id}/stock` (cached Bin per warehouse, as_of), `GET /parts/{id}/where-used`, `GET/POST /assets/{id}/bom`, `POST /work-orders/{id}/part-requests` (technician request; creates a hold), `POST /work-orders/{id}/issues`, `POST /work-orders/{id}/returns`, `POST /work-orders/{id}/kit`, `POST /transfers`, `POST /transfers/{id}/dispatch`, `POST /transfers/{id}/receive`, `GET /stock-matrix?part=` (all stores), `GET /stores/{id}/issue-queue`, `GET /critical-spares`, `GET/POST /reorder-proposals`, `GET /storerooms`
Repairs & workshops	`GET/POST /workshops`, `GET/POST /repair-jobs`, `POST /repair-jobs/{id}/transition` (dispatch, receive, repaired, scrap), `GET /workshops/{id}/queue`
Approvals	`GET /approvals?status=pending`, `POST /approvals/{id}/approve`, `POST /approvals/{id}/reject`, `POST /approvals/bulk`, `PUT /me/approval-delegate`
ERPNext integration	`POST /integrations/erpnext/webhook` (HMAC-verified), `GET /integrations/erpnext/health`, `GET /integrations/erpnext/outbox?status=`, `POST /integrations/erpnext/outbox/{id}/resend`, `POST /integrations/erpnext/outbox/{id}/cancel`, `POST /integrations/erpnext/reconcile` (admin), `POST /integrations/erpnext/pull?doctype=` (admin), `GET /integrations/erpnext/cursors`, `POST /work-orders/{id}/material-requests` (creates a pending approval)
Downtime	`GET/POST /downtime-events`, `PATCH /downtime-events/{id}`
Codes	`GET/POST /code-lists/{type}` (failure modes, mechanisms, causes, detection methods, actions, hold reasons, downtime types)
Schedule	`GET /schedule?site=&from=&to=`, `PATCH /schedule/{woId}`

Module	Representative endpoints
KPIs	`GET /kpi/pm-compliance`, `/kpi/mtbf`, `/kpi/mttr`, `/kpi/downtime`, `/kpi/backlog`, `/kpi/pareto`, `/kpi/site-comparison`. Common filters: site, area, class, asset, from, to
Notifications	`GET /notifications`, `POST /notifications/{id}/read`, `GET/PUT /me/notification-prefs`, `GET/POST /escalation-rules`
Documents	`POST /attachments` (multipart), `GET /attachments/{id}`
Users & RBAC	`GET/POST /users`, `PUT /users/{id}/site-roles`, `GET/POST /roles`, `GET /permissions`, `GET/POST /teams`
Audit	`GET /audit?entity=&id=&user=&from=&to=`
Sync	`POST /sync/push`, `GET /sync/pull`

Conventions:

- JSON bodies.
- ISO-8601 timestamps, stored in UTC and displayed in IST.
- Cursor pagination.
- `ETag/If-Match` for optimistic concurrency.
- RFC 7807 problem-details errors.
- `Accept-Language: ta|en`.

8.3 Key data entities (ERD outline)

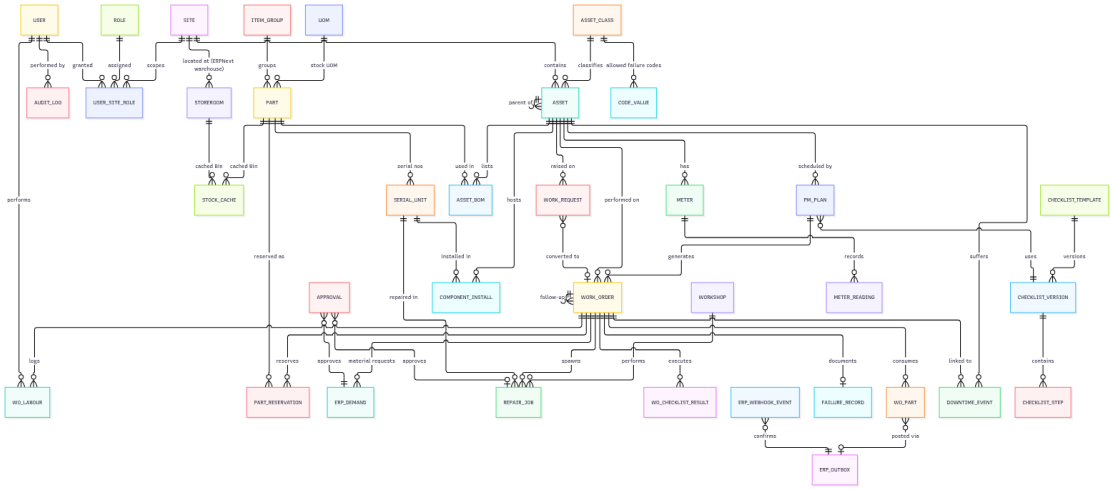


Figure 2 — Key data entities (ERD outline)

Entity	Key fields (outline)
`site`	id, code (MP / AP / CP), name_en, name_ta, timezone, wo_prefix, active
`asset`	id (UUID), site_id, parent_id, path (materialised path or closure table, for fast subtree queries), level_type, code, name_en, name_ta, asset_class_id, criticality, status, make, model, serial_no, commissioned_on, warranty_until, custom_fields (JSON), audit columns
`asset_class`	id, code, name_en/ta, custom_field_schema (JSON), default failure-code set
`asset_group`, `asset_group_member`	group of similar assets (may span sites) for PM/checklist application
`meter`, `meter_reading`	meter: asset_id, name, unit, type (cumulative/gauge), last_value. Reading: value, read_at, source (manual/checklist/import), user_id
`work_request`	id, site_id, asset_id, machine_state, symptom_code, description, requester_id, status, wo_id, created_at, client_uuid
`work_order`	id, number, site_id, asset_id, type, priority, status, repair_location, workshop_id, hold_reason, pm_plan_id, parent_wo_id, team_id, assignee_id, planned_start, due_at, started_at, completed_at, est_hours, actual_hours, cost_labour / cost_parts / cost_services, checklist_version_id, row_version (optimistic lock)
`wo_status_history`	wo_id, from_status, to_status, user_id, at, reason
`wo_labour`, `wo_part`	labour: user_id, start, end, hours. Part line: part_id, storeroom_id, direction (issue/return), qty, valuation_rate (from ERPNext), erp_stock_entry, outbox_id, status (pending/posted/failed/reversed)

Entity	Key fields (outline)
failure_record	wo_id, failed_component_asset_id, failure_mode_id, failure_mechanism_id, failure_cause_id, detection_method_id, failure_impact (critical/degraded/incipient), action_id, remarks
code_list / code_value	type (mode/mechanism/cause/detection/action/hold/downtime), code, label_en, label_ta, parent_id, asset-class scope
pm_plan	id, asset_id or asset_group_id, site_id, checklist_version_id, wo_type, trigger_type (time/meter/both/condition), interval_days, interval_meter, meter_id, schedule_mode (fixed/floating), lead_days, tolerance_pct, next_due_at, next_due_meter, active
checklist_template, checklist_version, checklist_step	template scope (global/site). Step: seq, type, text_en, text_ta, min, max, unit, photo_required, auto_followup, meter_id
wo_checklist_result	wo_id, step_id, value, pass_fail, photo_id, out_of_limit, user_id, at
downtime_event	asset_id, site_id, start_at, end_at, type, wo_id, request_id, source, corrected_by, correction_reason
part	erp_item_code (unique), mirrored: item_name, stock_uom, item_group, has_serial_no, has_batch_no, disabled, erp_modified. CMMS-owned: name_ta, photos, specs, mfr_part_no, criticality, is_critical_spare, alternates (JSON)
storeroom	code, erp_warehouse (unique), site_id (physical location), kind (master/sub/kitting/quarantine/workshop/vendor/transit), served_site_ids, is_default_for (site/area), parent_erp_warehouse, disabled, erp_modified
item_group, uom	Mirrored from ERPNext (name, parent, is_group / uom name, symbol), plus CMMS-owned name_ta; erp_modified cursor
workshop	code, name_en/ta, type (mechanical/electrical/other), site_id, team_id, wip_storeroom_id
repair_job	number, parent_wo_id, asset_id / component, serial_unit_id (nullable), repair_location (in_situ/mech_shop/elec_shop/external), workshop_id, vendor (ERPNext Supplier), status, estimate, approval_id, dispatched_at, expected_return_at, returned_at, outcome (repaired/scrapped), cost_labour/parts/service, erp_mr, erp_po
approval	subject_type (purchase_demand/external_repair/reorder_proposal/scrap), subject_id, requested_by, approver_id, delegate_of, status (pending/approved/rejected), reason, decided_at
sync_cursor	doctype, last_modified, last_full_sync_at, status
stock_cache	part_id, storeroom_id, actual_qty, projected_qty, ordered_qty, indented_qty, valuation_rate, as_of
part_reservation (demand hold)	wo_id or repair_job_id, part_id, storeroom_id, qty, status (requested/held/kitted/issued/released), requested_by (technician), issued_by (storekeeper), expires_at. Not a stock ledger.
asset_bom	asset_id or asset_class_id, part_id, qty, position/notes
erp_demand	wo_id or repair_job_id, type (purchase/transfer), part_id or free-text, qty, approval_id, erp_material_request, mr_status, po_no, receipt_status
erp_outbox	txn_id (UUID, unique), doctype, payload (JSON), status, attempts, next_attempt_at, last_error, erp_docname, created_at
erp_webhook_event	doctype, docname, event, modified, payload hash, processed_at (dedupe key = doctype + docname + event + modified)
serial_unit, component_install	serial_unit: part_id, serial_no (ERPNext Serial No), state (in store/installed/in repair/scrapped). Install: serial_unit_id, asset_id, installed_at, removed_at, wo_id
reorder_proposal	part_id, storeroom_id, proposed_level, proposed_qty, basis (JSON), status, approved_by, pushed_at
user, role, permission, user_site_role, team	user: emp_code, name, mobile, email, language, password_hash, active. Team: site_id, trade
attachment	owner_type, owner_id, file_key, mime, size, uploaded_by
notification, escalation_rule	escalation_rule: site_id, priority, threshold_minutes, target role/user
audit_log	entity, entity_id, action, changed_fields (JSON old/new), user_id, source, ip, at. Append-only; partitioned by month
kpi_daily	site_id, asset_id, date, operating_hours, downtime_min, failures, repairs, repair_min, pm_due, pm_on_time (pre-aggregated for dashboards)

MariaDB notes:

- InnoDB tables with utf8mb4 and a Unicode (UCA-based) collation that handles Tamil text.

- JSON columns for custom fields.
- Composite indexes such as `(site_id, status, due_at)` on work orders and `(asset_id, start_at)` on downtime.
- Partition `audit_log` and `meter_reading` by month as volumes grow.

8.4 Deployment architecture (on-premise, Ansible)

The system is **self-hosted on-premise** and deployed through the existing **PyCharm → Ansible** pipeline. The server location (e.g., the Main Plant server room) and the exact host layout are **[ASSUMPTION — validate]**.

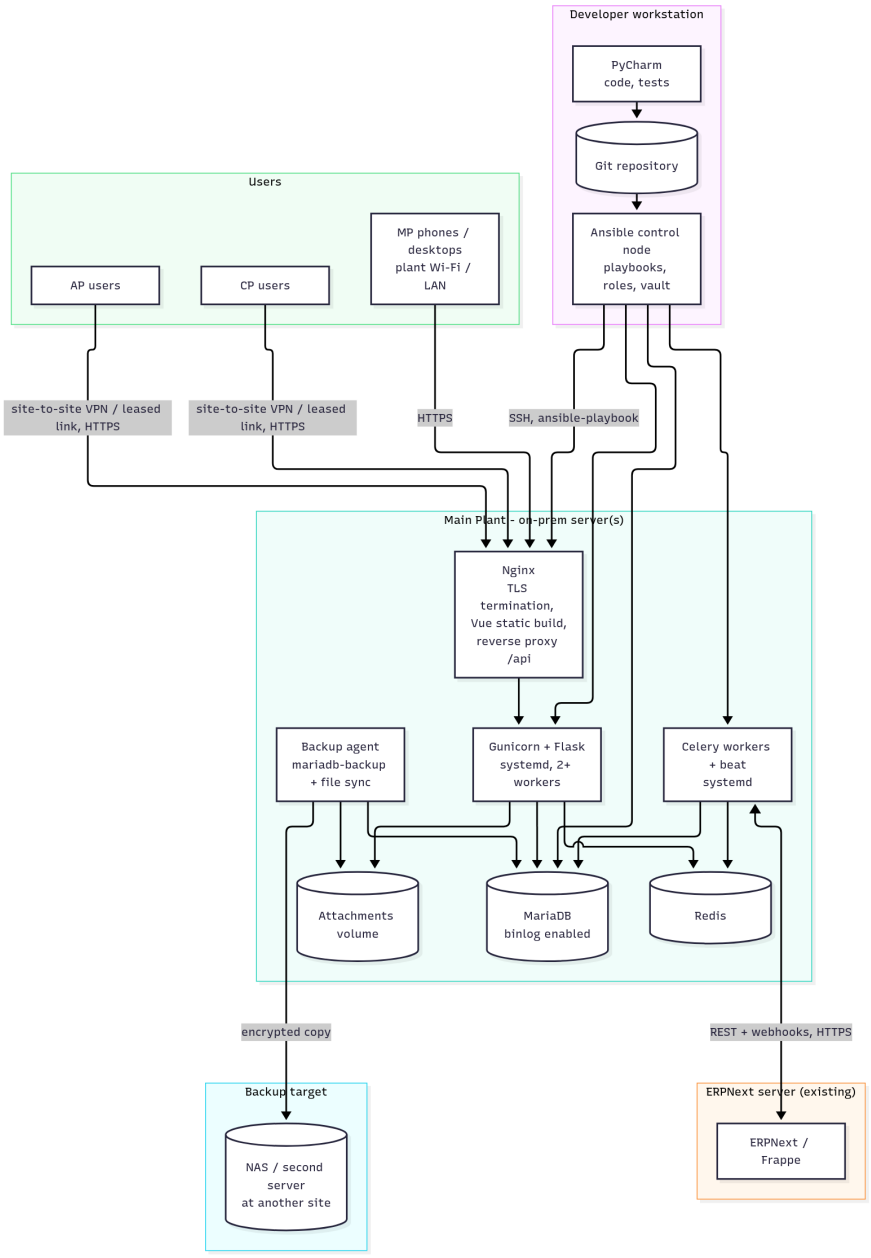


Figure 3 — On-premise deployment architecture (Ansible)

Ansible roles (proposed)

Standard role layout: `tasks/`, `handlers/`, `templates/`, `files/`, `vars/`, `defaults/`, `meta/` [S72]. Secrets go in Ansible Vault [S73].

Role	Responsibility
`common`	OS baseline, users/SSH keys, time sync (IST), firewall (allow 443 from site subnets/VPN only), fail2ban, unattended security updates

Role	Responsibility
`certs`	Install the TLS certificate/key for `cmms.<company-domain>` and the renewal hook. Distribute the internal CA bundle if an internal CA is used (see below)
`nginx`	Reverse proxy, TLS 1.2+/HSTS, gzip/brotli, caching headers for the hashed Vue assets, `no-cache` for `service-worker.js` and `index.html`, upload size limits
`mariadb`	Install/configure MariaDB, `utf8mb4`, InnoDB tuning, binary logging for PITR, app user with least privilege, slow query log
`redis`	Redis for Celery and cache, bound to localhost/private interface
`cmms_app`	Python venv, wheel/requirements install, `.env` from vault, Alembic migrations (`flask db upgrade`) run once per release, Gunicorn systemd unit, health check
`cmms_worker`	Celery worker and beat systemd units (PM generation, escalations, notifications, ERP sync, reconciliation)
`cmms_frontend`	Build or copy the Vue production bundle (`dist/`) to the Nginx web root, with atomic symlink switch
`erpnext_setup`	Idempotent fixture: custom fields, Stock Entry Types, warehouses (if agreed) and webhook definitions on ERPNext through its REST API, using vault secrets
`backup`	`mariadb-backup` full + incremental schedule, binlog retention, attachment sync, encrypted off-host copy, restore-test playbook
`monitoring`	Node/app metrics, log rotation, disk/backup alerts, uptime checks for `/api/health` and ERPNext sync age

Release flow:

1. Develop in PyCharm and run tests locally (pytest/Vitest).
2. Commit and tag a release in Git.
3. `ansible-playbook site.yml -l prod --tags release` deploys the new backend and frontend into versioned directories (`/opt/cmms/releases/<tag>`), runs migrations and restarts services.
4. A smoke check runs; on failure the `current` symlink is rolled back.

It is also recommended to add a CI test stage (e.g., a self-hosted Git runner) before the Ansible step, plus a **staging** inventory that mirrors production with an ERPNext test site.

HTTPS and certificates for the PWA

Service workers, and therefore offline mode and installability, need a **secure context**: HTTPS, with localhost exempt only for development [S74, S75]. Plain-HTTP intranet hosting will **not** work for the offline PWA. Options:

Option	How	Pros	Cons
1. Publicly trusted certificate for an internal hostname (recommended)	Use a company-owned domain (e.g., `cmms.<company-domain>`) resolved to the on-prem IP by internal/split DNS. Obtain a Let's Encrypt certificate with the **DNS-01** challenge, which does not need the server to be reachable from the internet; only the DNS TXT record must be public [S76]. Ansible renews it automatically.	Works on every phone (including personal) with no CA installation	Needs a domain with DNS API access; certificate renewal every 90 days (automated)
2. Internal CA	Company root CA (e.g., OpenSSL or step-ca), server certificate signed by it, root CA installed on all devices	Fully offline / no public DNS	Every phone must install the root CA. On Android 7+ apps don't trust user-installed CAs by default; Chrome is an exception [S77]. Hard to manage on personal phones
3. Self-signed	—	—	Not acceptable: browser warnings, PWA/offline unreliable

Access from the Ancillary Plant and Campus

- Site-to-site VPN (or an existing leased link/MPLS) between the three sites and the server. The firewall allows 443 only from site subnets and the VPN.
- Users at all sites use the same hostname via internal DNS.
- If the link drops, AP/CP technicians continue in PWA offline mode and sync later.
- Remote access for engineers outside the plants (if wanted) goes through a client VPN, not public exposure **[ASSUMPTION — validate whether off-site access is needed]**.

9. Spare parts & ERPNext integration

ERPNext **v16** is already used for purchase orders and inventory, and the user has confirmed that **inventory stays in ERPNext**: the CMMS does not keep its own stock ledger. The CMMS therefore works *with* ERPNext's stock ledger. It **pulls** master data and stock levels from ERPNext (read-mostly) and **writes back** only maintenance demand and movement documents (Material Requests and Stock Entries). A test ERPNext instance will be provided for development and contract testing. This section supersedes the earlier generic inventory (FR-INV) and procurement (FR-PRC) requirements.

9.1 ERPNext v16 capabilities relevant to maintenance spares (research)

ERPNext/Frappe **v16** was released on 12 January 2026 [S79, S80]. The facts below were checked against the v16 documentation or the `version-16` branch DocType definitions where possible [S82, S83, S84]. Items marked *(v15 doc)* come from earlier documentation and are expected to be unchanged, but should be confirmed on the test instance.

ERPNext / Frappe capability	What the sources say	Relevance to CMMS
REST API (Frappe)	Every DocType gets REST endpoints: `GET/POST /api/resource/:doctype`, `GET/PUT/DELETE /api/resource/:doctype/:name`, with `fields`, `filters`, `or_filters`, `order_by`, `limit_start/limit_page_length`, and an `expand` parameter for link fields. Whitelisted methods via `/api/method/<dotted.path>`. Auth: token (`Authorization: token api_key:api_secret`), session, or OAuth bearer. Calls run with the API user's roles [S50]. Frappe's integration guide recommends a dedicated least-privilege integration user, external IDs in custom fields, idempotent design and queues/webhooks [S88].	The CMMS pulls Items, Item Groups, UOMs, Warehouses and Bins and creates Material Requests and Stock Entries. No ERPNext app code is needed beyond custom fields and Stock Entry Types.
Submitting documents	Create then submit with `{"run_method":"submit"}`, or include `"docstatus": 1` on create; API v2 has `/api/v2/document/{doctype}/{name}/method/submit` [S51] *(v15 doc)*.	Stock Entries and Material Requests only affect stock or planning once submitted.
Item (v16)	Item fields include `item_group`, `stock_uom`, `uoms` (UOM conversion table), `disabled`, `is_stock_item`, `has_serial_no`, `has_batch_no`, `safety_stock`, `lead_time_days`, `reorder_levels`, `default_material_request_type`, `allow_alternative_item`, `default_manufacturer_part_no`, and a per-item `allow_negative_stock` flag [S83].	Item = the part master (`item_code` is the key). Mirrored read-only into the CMMS together with UOM conversions and item group.
Item Reorder (v16)	One row per warehouse: `warehouse_group`, `warehouse`, `warehouse_reorder_level`, `warehouse_reorder_qty`, `material_request_type` = **Purchase / Transfer / Material Issue / Manufacture** [S83]. When auto-reorder is enabled, a scheduled job raises Material Requests when projected qty falls below the level [S52, S53].	The **master store** reorders with type *Purchase*; **sub-stores** can be replenished automatically from the master store with type *Transfer*. ERPNext stays the only reorder engine.
Warehouse (v16)	Tree (`parent_warehouse`, `is_group`), per Company, with `warehouse_type` (e.g., Transit), `default_in_transit_warehouse`, `disabled`, `is_rejected_warehouse` [S83]. In v16 the Warehouse tree hides disabled warehouses by default [S79].	The CMMS mirrors the warehouse tree and maps each leaf to a CMMS `storeroom` (master store, sub-store, kitting, quarantine, workshop, vendor, transit).
UOM / Item Group	v16 adds common codes/symbols to UOM records and shows translated UOM names in link fields [S79].	Pulled so that the CMMS shows the same units and groups; a Tamil UOM label can be kept CMMS-side.
Bin (v16)	Per Item × Warehouse: `actual_qty`, `reserved_qty`, `ordered_qty`, `indented_qty`, `planned_qty`, `projected_qty`, `reserved_qty_for_production`, `reserved_qty_for_sub_contract`, `reserved_qty_for_production_plan`, `reserved_stock`, `valuation_rate`, `stock_value` [S83].	Cached in the CMMS. Available-to-promise subtracts ERPNext `reserved_stock` as well as CMMS holds.
Stock Entry (v16)	Purposes: Material Issue, Material Receipt, Material Transfer,	Maintenance movements use custom types

ERPNext / Frappe capability	What the sources say	Relevance to CMMS
	Material Transfer for Manufacture, Material Consumption for Manufacture, Manufacture, Repack, **Send to Subcontractor**, Disassemble, Receive from Customer, Return Raw Material to Customer, Subcontracting Delivery, Subcontracting Return [S83]. Custom Stock Entry Types map a new name onto a purpose [S58]; *Add to Transit* supports goods in transit [S59].	(see §9.4): **Maintenance Issue**, **Maintenance Return**, **Store Transfer**, **Kit to WO / Kit Return**, **Repairable Receipt**, **To Workshop / From Workshop**, **To Vendor for Repair / From Vendor Repair**.
Material Request (v16)	Types: Purchase, Material Transfer, Material Issue, Manufacture, Subcontracting, Customer Provided [S83, S60].	CMMS raises **MR (Purchase)** for shortages (after Admin approval) and **MR (Material Transfer)** for inter-store replenishment when the source store must fulfil it in ERPNext.
Purchase Order / Purchase Receipt	PO = supplier commitment, no ledger effect; Purchase Receipt posts stock (and accruals under perpetual inventory) [S62].	POs stay in ERPNext and are approved there by the Admin. The CMMS tracks PO/receipt status on the WO.
Stock Reservation (v16)	v16 extends reservation from Sales Orders to **Work Orders, Production Plans, Subcontracting Orders and Subcontracting Inward Orders**; adds **Auto Reserve Stock** in Stock Settings; reservations can originate from Pick List, Purchase Receipt, Stock Entry, Work Order, Production Plan [S79, S81, S82]. The `version-16` Stock Reservation Entry `voucher_type` options are exactly: Sales Order, Work Order, Subcontracting Inward Order, Production Plan, Subcontracting Order [S82].	Still **no maintenance document type** can own a reservation (ERPNext "Work Order" is the *manufacturing* work order). The CMMS therefore keeps **demand holds** (not a ledger) and uses the Kitting warehouse for physical hard reservation. If ERPNext Auto Reserve is turned on for sales/production, the CMMS must subtract `reserved_stock` from availability.
Webhooks (Frappe v16)	Per DocType and event: `after_insert`, `on_update`, `on_submit`, `on_cancel`, `on_trash`, `on_update_after_submit`, `on_change`, `workflow_transition`; optional condition, headers, JSON (Jinja) or form body; `timeout` (default 5 s); `background_jobs_queue`; HMAC-SHA256 signature in `X-Frappe-Webhook-Signature` when a secret is set; deliveries logged in *Webhook Request Log* [S65, S84]. The `version-16` Webhook DocType has **no retry-count field** (a `max_retries` field exists only on the develop branch at the time of research) [S84, S66].	Webhooks are a **fast hint**, not a guarantee. The CMMS must answer within the 5 s timeout (queue and return 200), and the reconciliation pull is mandatory because failed deliveries are not retried by v16.
Subcontracting (v16)	PO for a service item → **Subcontracting Order** → *Send to Subcontractor* Stock Entry to the supplier's warehouse → **Subcontracting Receipt**. v16 can reserve the raw materials on the Subcontracting Order [S85]. India Compliance can generate e-Waybills (Delivery Challan) for *Send to Subcontractor* Stock Entries and supports ITC-04 reporting with challan references [S86, S87].	A candidate for **external (vendor) repair** with GST job-work documentation (Option R2 in §9.5). Requires a subcontracting BOM per repairable item.
Custom Fields	Added via Customize Form / Custom Field; `custom_` prefix [S67].	`custom_cmms_wo`, `custom_cmms_asset`, `custom_cmms_site`, `custom_cmms_txn_id`, `custom_cmms_repair_job` on Stock Entry (+ detail) and Material Request.
Asset Maintenance / Asset Repair	Asset Maintenance: plans, tasks, teams, logs. Asset Repair: failure date, cost, actions, consumed stock (Material Issue), optional capitalisation [S68–S71] *(v15 doc)*.	**Overlap.** Too basic for plant maintenance (no requests, failure codes, downtime, meters, offline/QR, KPIs). **Do not use** for plant maintenance; optionally link CMMS assets to ERPNext fixed-asset records.

9.2 System-of-record options

Criterion	Option A: ERPNext is stock & valuation master (CMMS = demand & consumption source)	Option B: CMMS is stock master (ERPNext only for POs)	Option C: Hybrid (CMMS runs maintenance stores; ERPNext gets periodic summaries)
Stock ledger	One ledger (ERPNext)	Two ledgers; ERPNext Purchase Receipts still add stock in ERPNext, so they diverge immediately	Two ledgers, reconciled periodically
Valuation / accounting	ERPNext valuation rate and perpetual inventory stay correct	CMMS must replicate valuation	ERPNext value lags; adjustment entries needed

Criterion	Option A: ERPNext is stock & valuation master (CMMS = demand & consumption source)	Option B: CMMS is stock master (ERPNext only for POs)	Option C: Hybrid (CMMS runs maintenance stores; ERPNext gets periodic summaries)
Purchasing	Native: MR → PO → Receipt in ERPNext	Risk of double counting	Mixed
Effort to build	Medium: pull sync, cache, outbox, webhooks	High	High
Offline / mobile	Requests and used-qty queued; storekeeper posts issue on sync	Simple (local)	Simple (local)
Dependency risk	Mitigated by outbox + cache	Low	Medium
Data quality / audit	Single truth; every maintenance movement traceable to WO	Divergence likely	Periodic gaps
Fit with confirmed practice ("inventory managed in ERPNext")	**Best (confirmed)**	Contradicts it	Partial

Decision (confirmed by the user in v0.3): Option A. ERPNext owns items, UOMs, item groups, warehouses, quantities, valuation, purchasing and receipts. The CMMS **holds no stock ledger**; it holds a **read-mostly mirror** (items, groups, UOMs, warehouses, Bin cache) plus maintenance context:

- spares-to-asset BOM / where-used, criticality and critical-spare flags, Tamil names and photos;
- demand: part requests, holds, kits, shortages, approvals;
- consumption and returns against work orders, and repair jobs.

The CMMS **creates ERPNext documents** only for maintenance events: Stock Entries (issue, return, transfers, kitting, repair movements) and Material Requests (Purchase / Material Transfer). **Storekeepers** perform every issue; **technicians only request**. The **Admin** approves maintenance purchase demand.

9.3 Data ownership matrix

Data	Master	Copy / reference in the other system
Item code, name, item group, stock UOM, UOM conversions, serial/batch flags, disabled	ERPNext	CMMS `part`, `item_group`, `uom` mirror (read-only; pull sync)
Tamil name, photos, criticality, critical-spare flag, specs, alternates	CMMS	Optional push to ERPNext custom fields
Warehouse tree (master store, sub-stores, kitting, quarantine, workshop, vendor, transit)	ERPNext	CMMS `storeroom` mirror + kind/role mapping
Actual / projected / ordered / reserved qty, valuation rate	ERPNext (Bin)	CMMS `stock_cache` (pull + webhook-triggered refresh)
Reorder level / qty / MR type per warehouse	ERPNext Item Reorder	CMMS proposes; pushed after Admin approval
Part requests and holds for WOs / PM kits	CMMS (demand only, not stock)	Not in ERPNext (v16 reservations cannot reference a maintenance WO); hard hold = Kitting warehouse
Issue / return / transfers for maintenance	CMMS initiates (storekeeper) → ERPNext Stock Entry	CMMS stores the ERPNext document name
Purchase demand	CMMS (after Admin approval) → MR (Purchase)	ERPNext MR → PO (Admin approves in ERPNext) → Receipt
Spares-to-asset BOM / where-used	CMMS	Not in ERPNext
Repair jobs (location, workshop, vendor, cost, dates)	CMMS	ERPNext movements for serial units; service PO for external repair
Serial-numbered repairables: install/remove history	CMMS	ERPNext Serial No tracks warehouse location
Suppliers	ERPNext	CMMS reads supplier name for display / external repair

9.4 Store structure ↔ ERPNext warehouse mapping (revised v0.3)

Confirmed: one **master store** plus **a few sub-stores**, not necessarily one per site. **[ASSUMPTION — validate names, locations and which sites each sub-store serves against the ERPNext test instance]** Warehouse names below omit the ERPNext company suffix (e.g., " - SMTW").

```
<Company>
└── Maintenance Stores            (group)
    ├── Master Maintenance Store  (receipts; Purchase reorder)
    ├── Sub-store <name 1>        (e.g. AP; Transfer reorder)
    ├── Sub-store <name 2>        (e.g. CP or a plant area)
    ├── Kitting                   (PM kits, at master store)
    ├── Repair Quarantine         (failed repairables)
    ├── Mechanical Shop WIP       (under repair, mech shop)
    ├── Electrical Shop WIP       (under repair, elec shop)
    ├── At Vendor for Repair      (Option R1; or per vendor)
    └── Maintenance Transit       (type Transit; site moves)
```

CMMS `storeroom.kind`	Holds	Counted as "available"	Reorder in ERPNext
`master`	Main stock; all Purchase Receipts land here	Yes	Item Reorder, type **Purchase**
`sub`	Fast-moving and critical spares near the point of use	Yes	Item Reorder, type **Transfer** (source = master)
`kitting`	Picked kits for specific WOs (hard hold)	No (shown as "kitted for WO-x")	Never
`quarantine`	Removed/failed repairable units	No	Never
`workshop`	Units being repaired in Mechanical / Electrical shop	No	Never
`vendor`	Units at an external repairer	No	Never
`transit`	Goods moving between sites	No (shown as "in transit")	Never

Rules

- Each storeroom has a physical `site_id` (where it is) and a **served-sites list**. Each site and area has a **default issuing store** (e.g., AP technicians default to the AP sub-store, with fallback to the master store).
- The CMMS shows **availability across all stores** on every part, BOM and WO screen: a matrix of store × available / kitted / in transit / in repair, with the `as_of` time.
- **Inter-store transfers** are ERPNext **Material Transfer** Stock Entries (type "Store Transfer"). Between sites they go via *Maintenance Transit* (send → receive), so stock is never shown in two places. A sub-store can also ask for replenishment through an MR (Material Transfer) that the master storekeeper fulfils.
- **Kitting and repair warehouses are central** (one each, at the master store) unless a sub-store is far away. Then an optional per-sub-store Kitting leaf can be added without design changes.
- Cost centre per site (if used in ERPNext) is set on Stock Entries by the CMMS from the WO's site.

Maintenance Stock Entry Types (fixture applied by Ansible)

Stock Entry Type	Purpose (v16)	From → To
Maintenance Issue	Material Issue	master/sub/kitting → (consumed on WO)
Maintenance Return	Material Receipt	→ issuing store (unused parts issued directly; at original valuation rate)
Store Transfer	Material Transfer	store → store (or → Transit → store)
Kit to WO / Kit Return	Material Transfer	store ↔ Kitting
Repairable Receipt	Material Receipt	→ Repair Quarantine (removed unit; valuation per finance policy)
To Workshop / From Workshop	Material Transfer	Quarantine ↔ Mechanical/Electrical Shop WIP; Shop WIP → store when repaired
To Vendor for Repair / From	Material Transfer	Quarantine ↔ At Vendor for Repair (Option R1)

Stock Entry Type	Purpose (v16)	From → To
Vendor Repair		
Repair Scrap	Material Issue	Quarantine/Shop WIP → (scrapped; Admin approval)

9.4.1 Issue to a work order: Material Issue vs Material Transfer (v0.4)

The user describes the current ERPNext document as a **"material issue voucher"** and thinks it is a Material Transfer. In ERPNext v16 these are two different Stock Entry purposes [S83, S57]:

- **Material Issue** has a source warehouse only. Stock leaves inventory and is booked as consumption, with the line's `expense_account` and `cost_center` (perpetual inventory) [S83, S62].
- **Material Transfer** moves stock from one warehouse to another. It stays in inventory at the same value, and no expense is booked.

Design (default, "Issue mode"):

Movement	ERPNext Stock Entry purpose	Key fields set by the CMMS
Storekeeper issue against a WO / repair job	**Material Issue** (type "Maintenance Issue")	`s_warehouse` = issuing store; `custom_cmms_wo`, `custom_cmms_asset`, `custom_cmms_site`, `custom_cmms_repair_job`; `cost_center` (site/department); `expense_account` (maintenance consumption account, optionally per asset class)
Return of unused parts	Material Receipt (type "Maintenance Return") at the original rate, with the same WO, cost centre and account	same references
Inter-store transfer (master ↔ sub-store, via Transit between sites)	**Material Transfer** ("Store Transfer")	`s_warehouse`, `t_warehouse`
PM kitting / kit return	**Material Transfer** ("Kit to WO" / "Kit Return")	store ↔ Kitting
Repairable moves (quarantine, workshop WIP, vendor)	**Material Transfer**	per §9.4

Alternative mapping ("Transfer-to-consumption mode") if stores currently book issues as a Material Transfer to a consumption or department warehouse (e.g., "Maintenance Dept - SMTW"):

- The CMMS posts the WO issue as a **Material Transfer** from the issuing store to the configured consumption warehouse, still with `custom_cmms_wo`. The consumption warehouse is excluded from availability.
- **Consequence:** the parts remain in inventory value in ERPNext until finance writes them off. That needs a periodic **Material Issue** (or Stock Reconciliation) from the consumption warehouse, which the CMMS can prepare as a month-end summary per cost centre. Until then, maintenance cost per WO is known in the CMMS (valuation rate × qty) but not yet expensed in ERPNext.
- Returns become Material Transfers back from the consumption warehouse to the store.
- **Recommendation:** use **Issue mode** (Material Issue) for maintenance consumption, because it books cost at the time of use against the right cost centre and avoids a "ghost" stock balance. Support Transfer-to-consumption mode as a **configuration switch** (FR-ERP-001) in case finance or stores must keep the current practice during transition. Confirm with stores and finance (Open question 1).

9.5 Repair locations, workshops and external repair (new v0.3)

Confirmed: repairs happen in four places: **In-situ** (on the machine), the **Mechanical shop**, the **Electrical shop**, and **External** (vendor/subcontract).

On the work order. Every WO carries `repair_location` = In-situ | Mechanical shop | Electrical shop | External. For shop or external work it also carries `workshop_id` or `vendor`. A typical breakdown has a **parent WO** on the machine (remove and refit, usually in-situ) and a **child repair job** for the removed component at the workshop or vendor. Each workshop has its own queue (phone + desktop), its own technicians/team and its own KPIs (turnaround time, backlog, repeat repairs).

Repair location	What moves in ERPNext	Notes
In-situ	Only parts issued/returned against the WO	Default for most breakdowns and PM
Mechanical / Electrical shop, non-stocked component (e.g., a machine's own motor removed, rewound and refitted)	Nothing for the component (it is part of the asset, not stock); parts used in the shop are issued to the **repair job**	Tracked in the CMMS as `component_install` removal/refit plus a shop repair job
Mechanical / Electrical shop, **stocked repairable** (serial-numbered rotable spare)	Repairable Receipt → Quarantine; To Workshop → Shop WIP; From Workshop → store (or Repair Scrap)	Gives a correct "where is unit SN-x" answer in ERPNext
External (vendor)	See options R1–R3 below	GST job-work / delivery-challan treatment to be confirmed with the accountant

External repair options

Aspect	R1: Transfer + service PO (recommended to start)	R2: ERPNext Subcontracting	R3: CMMS-only tracking
ERPNext documents	Stock Entry "To Vendor for Repair" (Material Transfer to *At Vendor for Repair*); **PO for a repair service item** (Admin approves); on return "From Vendor Repair" transfer + Purchase Receipt/Invoice for the service	PO (service item) → Subcontracting Order → *Send to Subcontractor* → Subcontracting Receipt [S85]	Service PO only; unit stays in Quarantine in ERPNext while the CMMS records "at vendor"
Location accuracy	Good (unit is in the vendor warehouse)	Good (supplier warehouse)	Poor in ERPNext
GST / e-Waybill / ITC-04	Manual delivery challan unless India Compliance supports the chosen Stock Entry purpose **[validate]**	India Compliance e-Waybill (Delivery Challan) for *Send to Subcontractor* and ITC-04 challan references [S86, S87]	Manual
Setup effort	Low: one warehouse + one service item per repair category	High: subcontracting BOM per repairable item (failed unit in → repaired unit out); valuation adds the service cost to the repaired unit	Lowest
Fit	Few external repairs per month, mixed vendors	Many/high-value external repairs, or where job-work compliance is mandatory	Very low volume only

Recommendation: start with **R1** (simple and uses standard Material Transfer + PO). Keep **R2** as the target if external repair volume or GST job-work reporting justifies it. Confirm with finance which of R1/R2 matches how goods sent for repair are documented today. The CMMS `repair_job` model is the same for both; only the ERPNext posting adapter changes.

9.6 Integration architecture and sync patterns

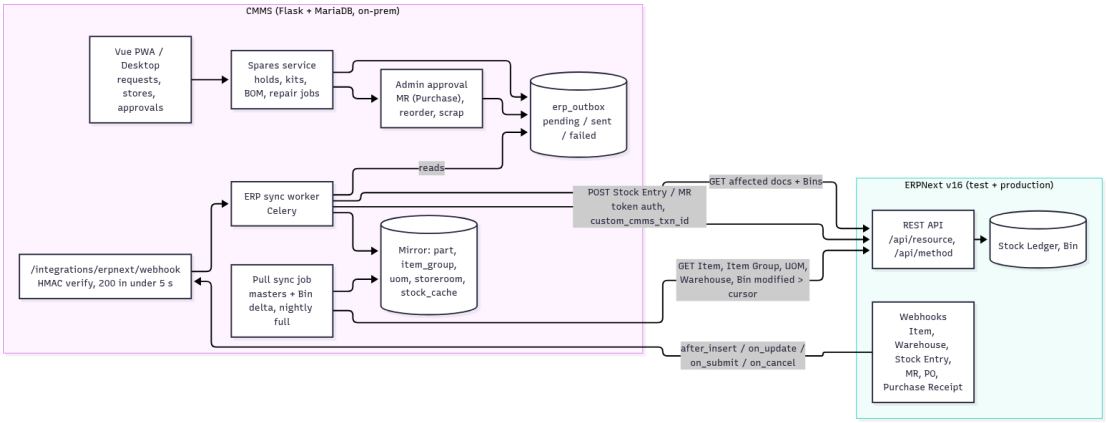


Figure 4 — CMMS ↔ ERPNext v16 integration architecture

Patterns

1. **Master-data pull (ERPNext → CMMS, read-mostly).** A Celery job pulls, in dependency order: **UOM → Item Group (tree) → Warehouse (tree) → Item (+ UOM conversions, reorder rows) → Bin → Supplier (name only) → Serial No (for repairable items)**.

- Each DocType keeps a `modified` cursor. Pages use `limit_start/limit_page_length` and `filters=[["modified",">",cursor]]` with `order_by=modified asc`.
- Items are limited to the configured maintenance item groups (e.g., "Maintenance Spares" and its children). Warehouses are limited to the "Maintenance Stores" group.
- Frequency: masters every 15 min (plus webhook hints) and a nightly full pass that also detects deletions/disablements. Bins: 15-min delta plus nightly full.
- Mirrored fields are **read-only** in the CMMS. CMMS-owned attributes live in separate columns and are never overwritten.

2. **Transactional outbox (CMMS → ERPNext writes).** Each write (issue, return, transfer, kit, repair movement, MR) is stored in `erp_outbox` in the same DB transaction as the business change: `txn_id`, doctype, payload, status, attempts, last_error, `erp_docname`. A user action never fails because ERPNext is slow or down.
3. **Idempotency.** `txn_id` → `custom_cmms_txn_id`. Before each retry, look up the existing document by that field, so retries never post twice.
4. **Retry and exceptions queue.** Exponential backoff for network/5xx errors. Business errors (insufficient stock, disabled item, closed period, negative-stock rule) go to the **ERP exceptions queue** for the storekeeper/Admin with resend / edit / cancel actions.
5. **Webhooks (hints).** On Item, Item Group, Warehouse (`after_insert`, `on_update`, `on_trash`) and on Stock Entry, Material Request, Purchase Order, Purchase Receipt (`on_submit`, `on_cancel`):
   - verify `X-Frappe-Webhook-Signature`;
   - store the event (dedupe on doctype + name + event + modified);
   - return 200 within ERPNext's 5 s default timeout;
   - a worker re-fetches the document and affected Bins.

   v16 webhooks have **no built-in retry count** [S84], and Bin updates are not relied on as webhook sources, so **pull reconciliation is the safety net**.
6. **Consumption reconciliation (nightly).** Compare CMMS WO part lines and repair movements with submitted ERPNext Stock Entries (by `custom_cmms_wo` / `custom_cmms_txn_id`). Raise mismatches (missing, cancelled in ERPNext, qty difference) to the exceptions queue.
7. **Availability semantics.** Per store: **Available to promise** = cached `actual_qty` − ERPNext `reserved_stock` − CMMS open holds − pending outbox issues/transfers-out. Kitting, quarantine, workshop, vendor and transit warehouses are shown separately, never as available. The UI always shows `as_of`; offline devices show "offline – may be stale".
8. **Security.** ERPNext user `cmms-integration`: read Item/Item Group/UOM/Warehouse/Bin/Supplier/Serial No/PO/Purchase Receipt; create + submit Stock Entry and Material Request only **[validate role set on the v16 test instance]**. Secrets in Ansible Vault; HTTPS only.
9. **Environments and contract tests.** The provided **ERPNext v16 test instance** is the integration target for development and CI contract tests. A setup fixture creates custom fields, Stock Entry Types, the maintenance warehouse group, test items and webhooks. Production gets the same fixture via Ansible after sign-off.

9.7 Key flows (sequence diagrams)

9.7.1 Request and issue parts against a work order (technician requests, storekeeper issues)

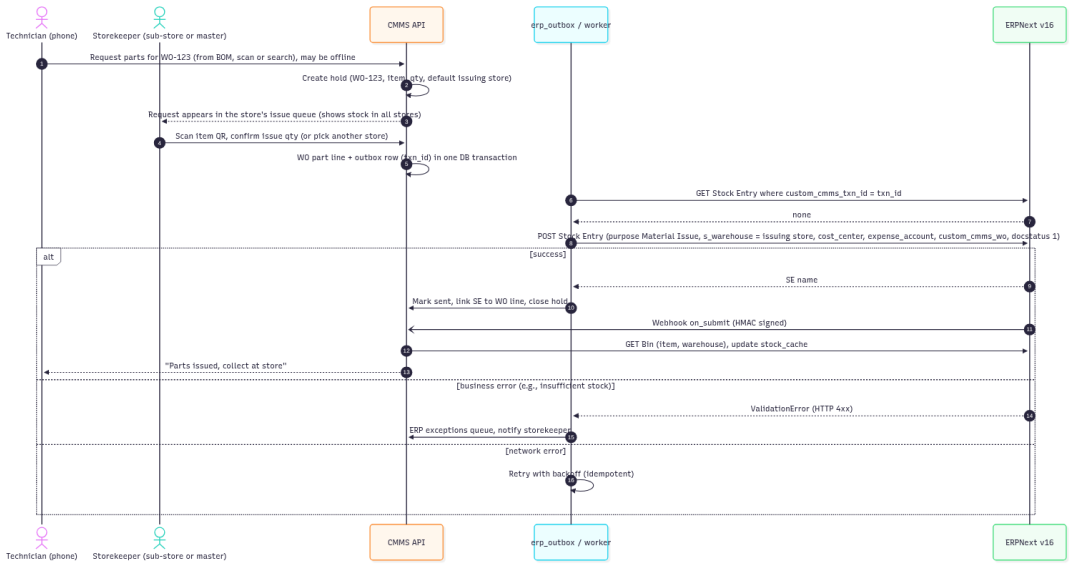


Figure 5 — Sequence: technician requests, storekeeper issues parts

9.7.2 Shortage → Admin approval → Material Request (Purchase) → PO → Receipt

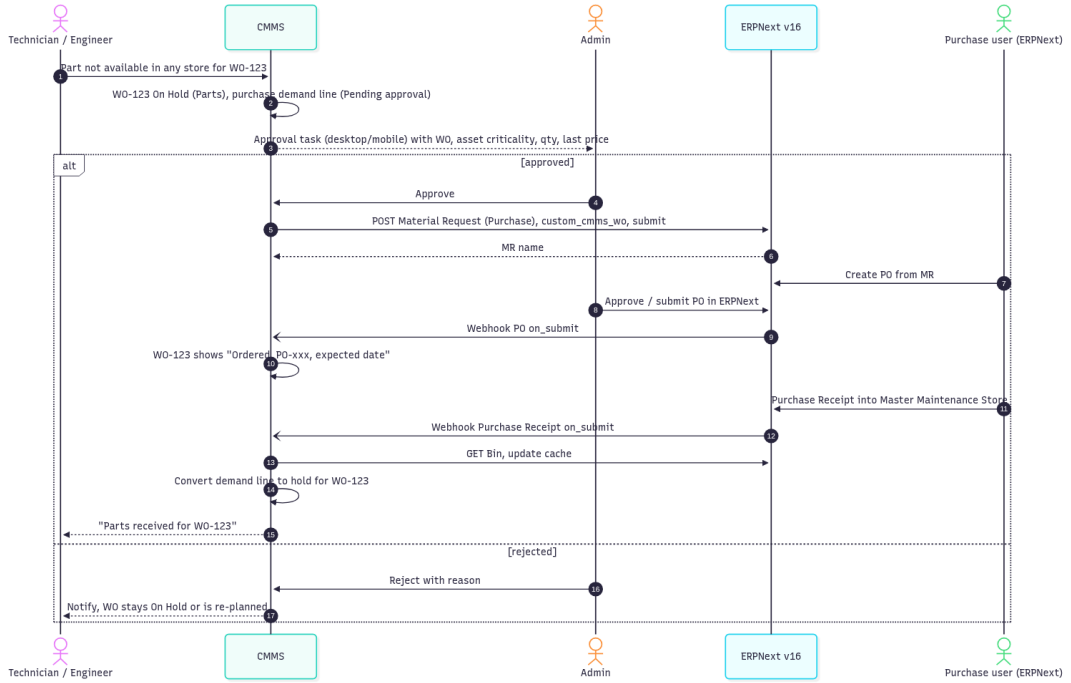


Figure 6 — Sequence: shortage → Admin approval → MR → PO → receipt

9.7.3 Inter-store transfer (master → sub-store, or between sub-stores)

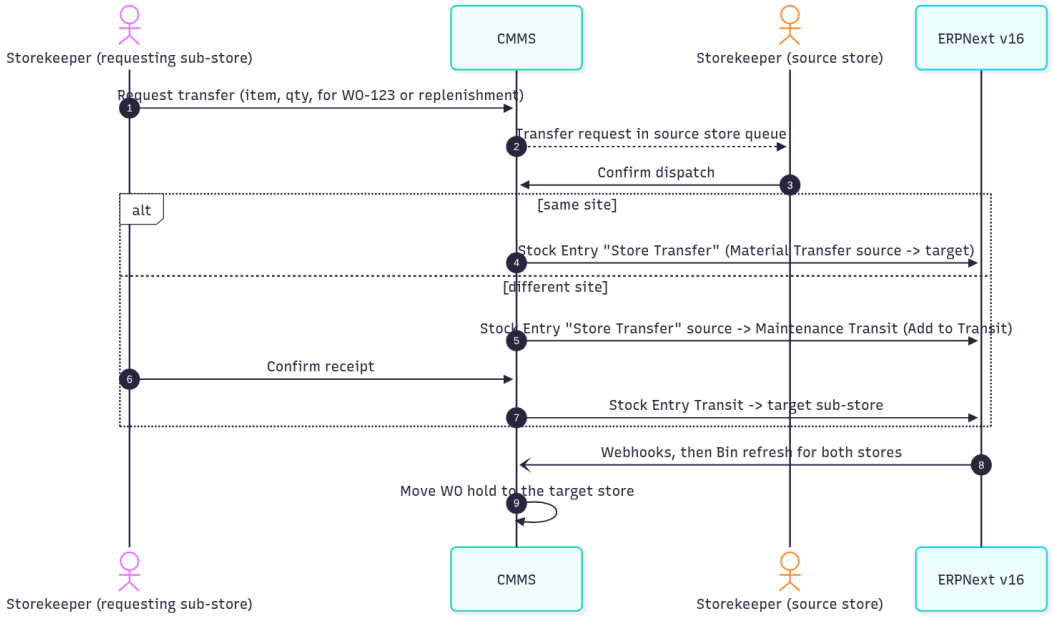


Figure 7 — Sequence: inter-store transfer

Sub-store replenishment can also be automatic: ERPNext Item Reorder with type Transfer raises an MR (Material Transfer), which the master storekeeper fulfils in the CMMS or ERPNext.

9.7.4 PM kitting and return of unused parts

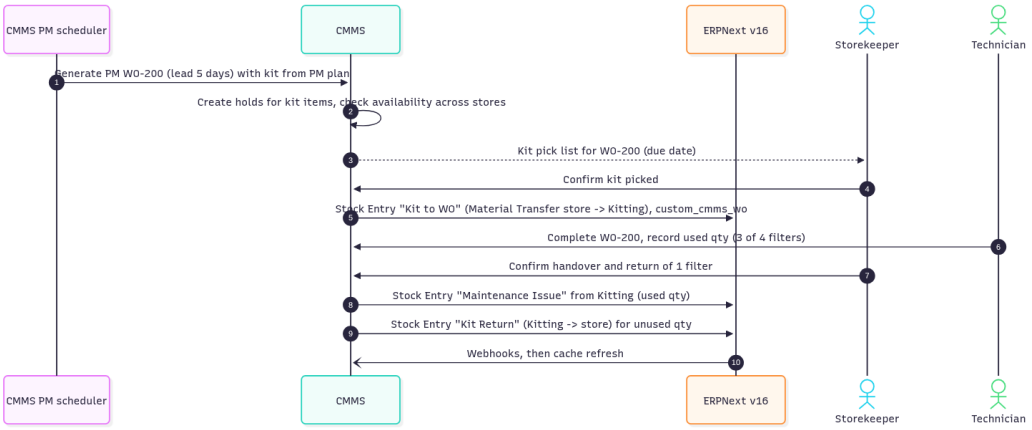


Figure 8 — Sequence: PM kitting and return of unused parts

9.7.5 Repairable spare via Mechanical / Electrical shop

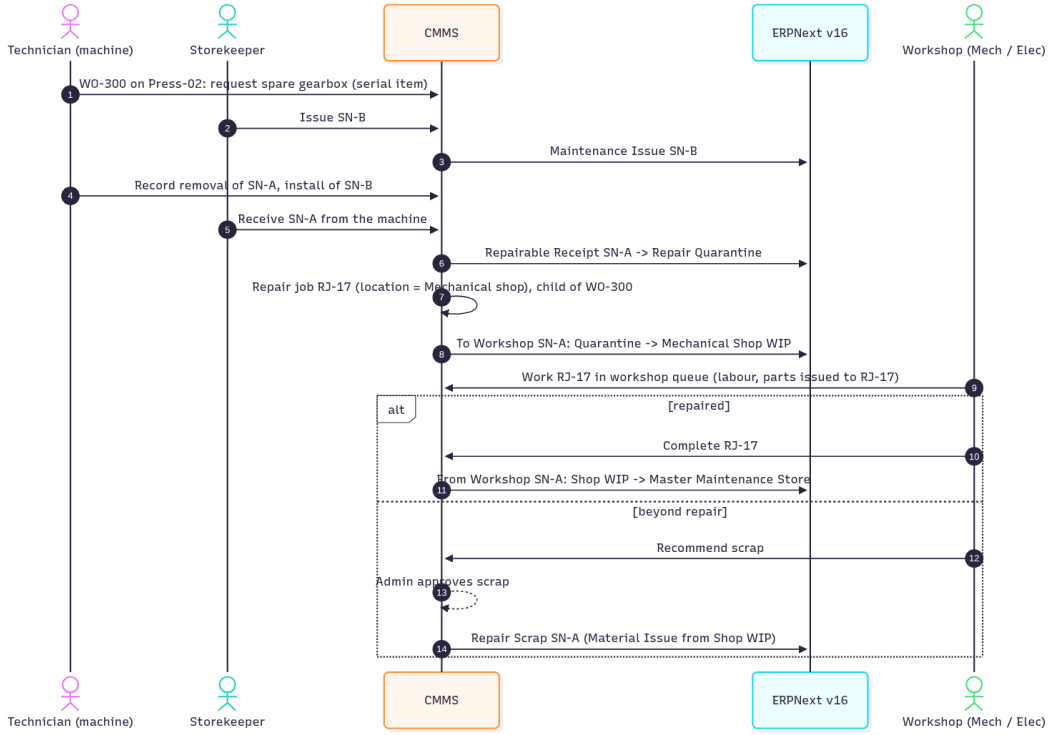


Figure 9 — Sequence: repairable spare via Mechanical / Electrical shop

9.7.6 External (vendor) repair: Option R1

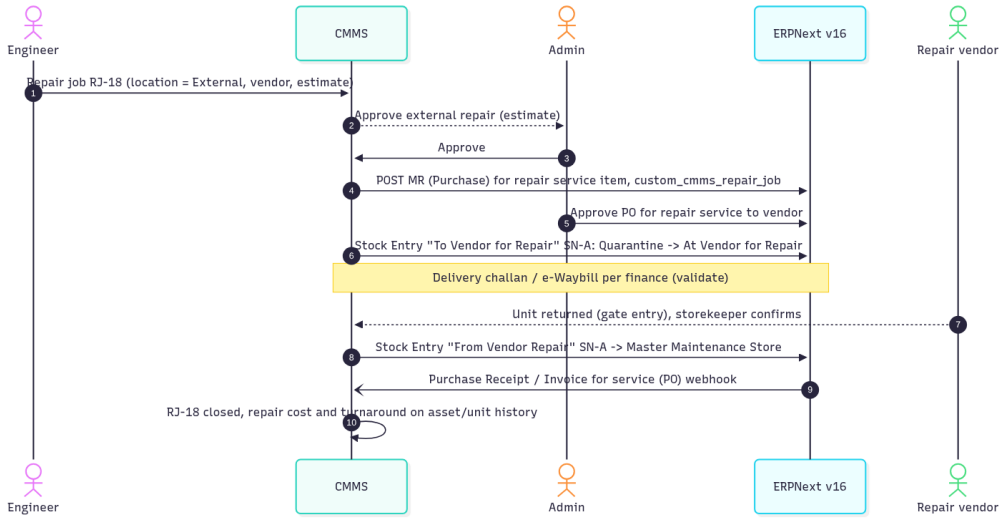


Figure 10 — Sequence: external (vendor) repair, Option R1

Option R2 replaces steps 5–9 with an ERPNext Subcontracting Order, a "Send to Subcontractor" Stock Entry and a Subcontracting Receipt [S85]. The CMMS repair job is unchanged.

The accounting treatment of repairables (valuation on receipt into quarantine, repair cost capitalisation or expense) must be agreed with finance; see Open Questions.

9.8 Functional requirements: spares (SPR) and ERPNext integration (ERP)

ID	Requirement	Pri	MVP
FR-SPR-001	The CMMS part master shall mirror ERPNext Items in the configured maintenance item group(s): item_code, name, item group, stock UOM, UOM conversions, serial/batch flags, disabled, reorder	M	✓

ID	Requirement	Pri	MVP
	rows. Mirrored fields are read-only in the CMMS.		
FR-SPR-002	CMMS-owned part attributes: Tamil name, photos, technical specification, manufacturer part no., criticality, critical-spare flag, interchangeable/alternate parts.	M	✓
FR-SPR-003	Storerooms shall mirror ERPNext leaf warehouses under the maintenance group. Each has a kind (master / sub / kitting / quarantine / workshop / vendor / transit), a physical site, served sites, and a default-issuing-store rule per site/area. A uniform per-site structure is not required.	M	✓
FR-SPR-004	Stock cache per item × warehouse (actual, reserved_stock, projected, ordered, requested qty, valuation rate, as_of), refreshed by webhook hints, a 15-minute delta pull and a nightly full pull. The CMMS shall hold **no stock ledger** of its own.	M	✓
FR-SPR-005	Show **availability across all stores** (store × available / kitted / in transit / in repair, with as_of) on WO, asset BOM and part screens. Available = actual − ERPNext reserved_stock − CMMS holds − pending outbox issues/transfers-out.	M	✓
FR-SPR-006	Spares-to-asset BOM: link parts to assets and/or asset classes with qty per position. Where-used view per part (which machines at which sites).	M	✓
FR-SPR-007	Technicians shall **request** parts for a WO or repair job (from the BOM, by scan or search), online or offline. Requests create holds and appear in the issuing store's queue. **Technicians cannot issue stock.**	M	✓
FR-SPR-008	**Only storekeepers** (store role, per storeroom) shall issue parts against a WO or repair job (scan item/bin QR, confirm qty, optionally switch store). The CMMS posts an ERPNext Stock Entry with purpose **Material Issue** (type "Maintenance Issue": consumption, source warehouse only). It carries `custom_cmms_wo`, asset, site and repair-job references, and sets `cost_center` and `expense_account` per line from the WO's site/asset class mapping. Inter-store, kitting and repair moves use **Material Transfer** (§9.4.1).	M	✓
FR-SPR-009	Technicians may record **used quantity** offline at WO completion. The difference between issued and used becomes a pending return for the storekeeper to confirm.	M	✓
FR-SPR-010	Returns: the technician initiates the return of unused parts; the storekeeper confirms physical receipt. The CMMS posts "Maintenance Return" or "Kit Return". The quantity returned cannot exceed the quantity issued to that WO.	M	✓
FR-SPR-011	PM kitting: PM plans list required parts. At WO generation (lead time) the CMMS creates holds and a pick list. On picking, it transfers the kit to the Kitting warehouse ("Kit to WO").	S	✓
FR-SPR-012	Shortage handling: when no store has enough available stock, the CMMS offers (a) an inter-store transfer from a store that has stock, or (b) a purchase demand line requiring **Admin approval** before an MR (Purchase) is posted. The WO goes On Hold – Parts.	M	✓
FR-SPR-013	Non-stock purchases for a WO (e.g., a special bearing) create a purchase demand line with free-text description, approved by the Admin, then posted as an MR (Purchase).	S	✓
FR-SPR-014	Track MR → PO → Receipt status on the WO (via webhooks/pull) and notify the WO owner when parts are received.	S	✓
FR-SPR-015	Critical spares: flag parts required for A-critical assets. Dashboard of critical spares below minimum, per store and in total. Alert when a critical spare reaches 0 available in all stores.	M	✓
FR-SPR-016	Min/max proposal: the CMMS shall suggest reorder level/qty per store (Purchase for master, Transfer for sub-stores) from consumption history, lead time and criticality. After **Admin approval**, push to ERPNext Item Reorder rows. ERPNext remains the only engine that auto-raises reorder MRs.	S	
FR-SPR-017	Repairable spares: serial-numbered items have install/remove history on assets; removed units go to Repair Quarantine; a **repair job** routes them to the Mechanical shop, Electrical shop or an external vendor, with matching ERPNext transfers (§9.4); repaired units return to store; scrap needs Admin approval.	S	
FR-SPR-018	Inter-store transfers (master ↔ sub-stores, sub ↔ sub) initiated by a storekeeper or from a WO shortage, posted as ERPNext Material Transfer ("Store Transfer"), via Maintenance Transit when between sites (two-step send/receive).	M	✓
FR-SPR-019	Spares consumption reports: per asset, asset class, site, store, workshop, failure mode and period, with value from the ERPNext valuation rate.	S	✓
FR-SPR-020	Greenfield onboarding: bulk import templates (Excel/CSV) for (a) new spare items to create in ERPNext (ERPNext Data Import), (b) CMMS part attributes, (c) asset BOM links, and (d) opening stock per store (posted in ERPNext as Stock Reconciliation by stores). Each template has a validation report.	M	✓
FR-SPR-021	Physical counts remain in ERPNext (Stock Reconciliation). The CMMS shows variances affecting maintenance items after the next sync.	C	
FR-SPR-022	Sub-store replenishment requests: a sub-store can raise an MR (Material Transfer) to the master store; ERPNext Item Reorder type Transfer can raise these automatically. The CMMS shows open replenishment MRs per store.	S	

ID	Requirement	Pri	MVP
FR-SPR-023	External repair (Option R1): repair job with vendor, estimate, Admin approval, MR (Purchase) for the repair service item, “To Vendor for Repair” / “From Vendor Repair” transfers, expected return date with overdue alerts, and a printable returnable delivery challan **[format to be validated with finance]**.	S	
FR-SPR-024	External repair (Option R2, optional adapter): post the same repair job as an ERPNext Subcontracting Order / Send to Subcontractor / Subcontracting Receipt instead of R1, selectable per repair category.	C	
FR-APR-001	**Single-step approval by the one Admin of the instance** (all sites) for: maintenance purchase demand (MR Purchase), external repair dispatch/estimate, reorder-level pushes, and repair scrap. No per-site approvers, second approver or amount-based threshold.	M	✓
FR-APR-002	Approval inbox for the Admin on desktop and mobile, with bulk approve, reject with reason, and full context (WO, asset criticality, stock in all stores, last purchase rate from ERPNext).	M	✓
FR-APR-003	Optional: the Admin may name a delegate approver for a date range (leave cover only). Delegated approvals are audited with both names, and the delegation ends automatically.	S	
FR-APR-004	Purchase Orders are created and approved in ERPNext (by the Admin’s ERPNext user). The CMMS does not approve POs itself; it displays PO status from ERPNext.	M	✓
FR-APR-005	The system shall allow exactly one active user with the Admin role per instance. Reassigning Admin is an audited handover (the old Admin loses the role).	M	✓
FR-ERP-001	The ERPNext connection (base URL, API key/secret, company, item groups, warehouse group and kinds, Stock Entry types, service items, issue mapping mode (§9.4.1), cost centre and maintenance expense account per site/asset class) shall be configurable by an admin per environment (test / production). Secrets are stored encrypted and deployed via Ansible Vault.	M	✓
FR-ERP-002	Transactional outbox for all CMMS → ERPNext writes, with statuses pending/sent/failed/cancelled, attempts, last_error and erp_docname.	M	✓
FR-ERP-003	Idempotent posting using `custom_cmms_txn_id`; look up the document before any retry.	M	✓
FR-ERP-004	Retry with exponential backoff for transient errors. Business errors go to an ERP exceptions queue with resend/edit/cancel actions and an audit trail.	M	✓
FR-ERP-005	Webhook receiver for Item, Item Group, Warehouse (`after_insert`, `on_update`, `on_trash`) and Stock Entry, Material Request, Purchase Order, Purchase Receipt (`on_submit`, `on_cancel`). HMAC-SHA256 verification, deduplication, 200 response within 5 s, asynchronous processing.	M	✓
FR-ERP-006	Scheduled pull and reconciliation: masters (UOM, Item Group, Warehouse, Item, Supplier, Serial No) and Bin deltas every 15 min by `modified` cursor; nightly full pull with deletion/disable detection; nightly WO-consumption vs Stock Entry comparison. Discrepancies go to the exceptions queue.	M	✓
FR-ERP-007	Integration health dashboard: last successful pull per DocType, outbox depth, failed items, webhook lag. Alerts the admin when sync is older than a threshold (e.g., 30 min).	S	✓
FR-ERP-008	A cancellation in ERPNext of a CMMS-originated Stock Entry shall flag the WO part line as “reversed in ERPNext” and notify the storekeeper.	S	✓
FR-ERP-009	Required ERPNext custom fields, Stock Entry Types, maintenance warehouse group/leaves, service items and webhooks shall be delivered as a versioned setup fixture, applied first to the **ERPNext v16 test instance** and then to production by Ansible.	S	✓
FR-ERP-010	Optional link of CMMS assets to ERPNext Asset records (fixed-asset accounting). No PM scheduling in ERPNext Asset Maintenance.	C	
FR-ERP-011	Tally integration.	W	
FR-ERP-012	Contract test suite against the ERPNext v16 test instance (create/submit/cancel Stock Entry and MR, pull masters, webhook signature) runs in CI before each release.	S	✓

9.9 Additional use cases

UC-13 Request and issue parts against a WO

- **Actors:** Technician (requests), Storekeeper (issues).
- **Preconditions:** WO is in progress; items are mirrored; ERPNext connection is configured.
- **Main flow:** see sequence 9.7.1.
- **Alternates:**
  -
    a) Technician is offline: the request is queued and reaches the store queue on sync.
  -

b) ERPNext rejects for insufficient stock: exceptions queue; the storekeeper reconciles or issues from another store.

c) Default store has no stock: the storekeeper issues from another store at the same site, or starts an inter-store transfer (UC-19).

UC-14 Handle a parts shortage

- **Actors:** Technician/Engineer, Admin.
- **Main flow:** see sequence 9.7.2.
- **Alternates:**
  - a) Stock exists in another store: transfer instead of purchase (UC-19).
  - b) Admin rejects: the requester is notified with the reason.
  - c) Urgent local purchase outside ERPNext: record a non-stock cost on the WO and flag it for an ERPNext entry.

UC-15 Kit parts for a PM

- **Actors:** System, Storekeeper, Technician.
- **Main flow:** see sequence 9.7.4.
- **Alternate:** Kit not picked by the due date: the planner is alerted and the WO is flagged “kit not ready”.

UC-16 Return unused parts

- **Actors:** Technician (initiates), Storekeeper (confirms).
- **Main flow:**
  1. At completion the technician records used qty; the difference becomes a pending return.
  2. The technician hands the parts to the store.
  3. The storekeeper confirms qty and condition.
  4. The CMMS posts “Maintenance Return” (or “Kit Return”); WO cost is reduced.
- **Alternate:** Damaged part returned: record as scrap with a reason; Admin approval.

UC-17 Repair a rotable component in a workshop

- **Actors:** Technician, Storekeeper, Workshop technician (Mechanical or Electrical), Admin (scrap only).
- **Main flow:** see sequence 9.7.5.
- **Alternates:**
  - a) Non-stocked component of the machine (e.g., its own motor): no ERPNext movement for the component; repair job only, then refit.
  - b) Beyond economic repair: scrap with Admin approval.
  - c) Workshop decides to send it out: convert to external repair (UC-20).

UC-18 Resolve ERP sync exceptions

- **Actor:** Storekeeper/Admin.
- **Main flow:** open the queue, review the ERPNext message and payload, fix and resend or cancel with a reason; the action is audited.
- **Alternate:** Nightly reconciliation mismatch: compare WO lines with ERPNext Stock Entries and choose which side to correct.

UC-19 Transfer stock between stores

- **Actors:** Storekeeper (requesting store), Storekeeper (source store).
- **Main flow:** see sequence 9.7.3.
- **Alternate:** Automatic replenishment MR (Transfer) from ERPNext Item Reorder appears in the master store queue.

UC-20 Send a unit for external repair

- **Actors:** Engineer, Admin, Storekeeper, Vendor.
- **Main flow:** see sequence 9.7.6 (Option R1).
- **Alternates:**
  - a) Vendor quote exceeds the approved estimate: re-approval by the Admin.
  - b) Unit overdue from the vendor: alert to the engineer and Admin.

UC-21 Approve maintenance demand (Admin)

- **Actor:** the Admin (single approver for all sites), or an optional leave-cover delegate.
- **Main flow:**
  1. Open the approval inbox (desktop or phone).
  2. Review the item: purchase demand, external repair, reorder proposal or scrap.
  3. Approve or reject with a reason; bulk approve is allowed.
  4. Approved purchase demand is posted to ERPNext as an MR (Purchase); the PO is then approved by the Admin in ERPNext.
- **Alternate:** Admin on leave: if a delegate is configured (optional), the delegate approves and both names are recorded; otherwise items wait in the inbox with an ageing alert.

9.10 Additional user stories (Epic E10: Spares & ERPNext)

US-43 As a technician, I want to see available stock in every store for the parts on my machine's BOM so that I know whether I can repair it now.

- **Given** Press-02 has BOM part "Seal kit", the master store has actual_qty 4 with 1 held in the CMMS, and the AP sub-store has 2, **When** I open the asset's spares tab, **Then** I see Master 3, AP sub-store 2, each with an "as of" time.

US-44 As a storekeeper, I want issuing parts in the CMMS to post a Stock Entry in ERPNext automatically so that I don't enter it twice.

- **Given** WO-123 requests 2 × Seal kit, **When** I confirm the issue from the master store, **Then** a submitted ERPNext Stock Entry "Maintenance Issue" (purpose Material Issue) for 2 units with `custom_cmms_wo = WO-123`, the MP cost centre and the maintenance expense account exists, **And** the WO line shows the SE number.

US-45 As a storekeeper, I want failed postings to go to a queue so that nothing is lost when ERPNext rejects an entry.

- **Given** ERPNext rejects an issue for insufficient stock, **When** the worker gets the error, **Then** the item appears in the ERP exceptions queue with the ERPNext message, **And** I am notified.

US-46 As the system, I want postings to be idempotent so that retries never create duplicate stock deductions.

- **Given** an issue was posted but the response timed out, **When** the worker retries, **Then** it finds the existing SE by `custom_cmms_txn_id`, **And** does not create a second one.

US-47 As a planner, I want PM kits staged before the due date so that technicians don't wait at the store.

- **Given** PM WO-200 is generated 5 days before due with a 4-item kit, **When** the storekeeper confirms picking, **Then** a "Kit to WO" transfer to Kitting is posted, **And** the WO shows "Kit ready".

US-48 As a technician, I want to return unused parts against the WO so that stock and WO cost are correct.

- **Given** 4 filters were issued to WO-200 and I recorded 3 used, **When** the storekeeper confirms the return of 1, **Then** a "Kit Return" is posted, **And** the WO parts cost reflects 3 filters.

US-49 As an engineer, I want a shortage to create a purchase request for Admin approval so that purchasing knows why it is needed.

- **Given** no store has the part for WO-123, **When** I choose "Request purchase", **Then** a demand line "Pending Admin approval" is created, **And** WO-123 is On Hold – Parts.

US-50 As a WO owner, I want to be notified when ordered parts arrive so that the job can resume.

- **Given** an approved MR linked to WO-123 became a PO, **When** a Purchase Receipt for the item is submitted into the master store, **Then** I get a notification, **And** the parts are held for WO-123.

US-51 As a maintenance manager, I want critical spares below minimum highlighted so that breakdown risk is controlled.

- **Given** "Kiln burner nozzle" is a critical spare with minimum 2 across stores, **When** the cache shows 1 available in total, **Then** it appears red on the critical-spares dashboard, **And** I receive an alert.

US-52 As an Admin, I want the CMMS to propose reorder levels per store so that ERPNext's auto-reorder is well tuned.

- **Given** 12 months of consumption and a 14-day lead time, **When** I approve a proposal, **Then** Item Reorder rows are written (type Purchase for the master store, Transfer for sub-stores).

US-53 As a planner, I want to see which machines use a part so that I can standardise spares across sites.

- **Given** a bearing is linked to 9 assets at MP and AP, **When** I open its where-used view, **Then** I see all 9 assets grouped by site.

US-54 As an admin, I want an integration health panel so that I notice sync problems early.

- **Given** the last successful Bin pull was 45 minutes ago (threshold 30), **When** I open the admin dashboard, **Then** the ERPNext status is amber with outbox depth and last error, **And** an alert has been sent.

US-55 As a technician, I want to request parts but not issue them so that stores control is kept.

- **Given** I hold the Technician role, **When** I open a WO's parts tab, **Then** I can create requests and record used qty, **And** no "Issue" action is available to me.

US-56 As the Admin, I want one approval inbox so that maintenance demand is approved quickly.

- **Given** 3 purchase demands and 1 scrap request are pending, **When** I open the inbox on my phone, **Then** I see all 4 with WO, asset criticality and stock in all stores, **And** I can approve or reject each with a reason.

US-57 As a storekeeper, I want to transfer stock to a sub-store at another site so that it is available where needed.

- **Given** the master store has 6 V-belts and the AP sub-store has 0, **When** I dispatch 2 to AP, **Then** a "Store Transfer" to Maintenance Transit is posted, **And** after the AP storekeeper confirms receipt, a transfer Transit → AP sub-store is posted and AP shows 2.

US-58 As an engineer, I want each WO to record where the repair is done so that workshop workload and turnaround are visible.

- **Given** a gearbox is removed from Press-02 under WO-300, **When** I create a repair job with location "Mechanical shop", **Then** it appears in the Mechanical shop queue, linked to WO-300, **And** the unit moves to Mechanical Shop WIP in ERPNext (if it is a stocked serial item).

US-59 As a workshop technician, I want to complete or scrap a repair job so that the unit's status is correct.

- **Given** RJ-17 is in the Electrical shop, **When** I mark it repaired, **Then** a "From Workshop" transfer to the master store is posted, **And** the unit's history shows the repair labour and parts.

US-60 As an engineer, I want to send a unit to an external vendor with Admin approval so that external repairs are controlled and traceable.

- **Given** RJ-18 has an estimate of ₹18,000 from Vendor X, **When** the Admin approves, **Then** an MR (Purchase) for the repair service is posted, **And** after the storekeeper dispatches the unit, a "To Vendor for Repair" transfer exists, **And** an overdue alert fires if the unit is not back by the expected date.

US-61 As the system, I want to pull items, UOMs, item groups and warehouses from ERPNext so that the CMMS always uses ERPNext's masters.

- **Given** a new Item is created in ERPNext in the "Maintenance Spares" group, **When** the next pull or webhook is processed, **Then** the part appears in the CMMS with its UOM and group as read-only fields, **And** a disabled ERPNext item becomes unavailable for new requests.

10. Suggested MVP phasing / roadmap

Durations are indicative and assume a small team (e.g., 2–3 developers plus the architect) **[ASSUMPTION — validate team size]**.

Phase	Duration (indicative)	Scope	Exit criteria
0. Foundations	3–4 weeks	Repo; Ansible roles + staging and production inventories; TLS certificate (DNS-01) and VPN access from AP/CP verified; backups; auth/RBAC with site scoping; MP/AP/CP sites configured; i18n (EN/TA); audit framework; code lists; phone-first shell and desktop workspace shell; **ERPNext v16 test instance (provided)** with custom fields, Stock Entry Types, maintenance warehouses and webhooks fixture; integration user; ERPNext contract tests in CI	One-command Ansible deploy to staging and production; PWA installs on a technician phone over HTTPS from all three sites; restore test passes; master pull and Stock Entry/MR write pass contract tests on the ERPNext v16 test instance
1. MVP: pilot at the Main Plant	10–12 weeks	Asset registry and import (all ~500 machines across 3 sites can be loaded; pilot uses MP); QR labels; work requests; full WO lifecycle; failure codes; automatic downtime capture; **ERPNext spares v1**: pull of Items, UOMs, Item Groups, Warehouses and Bins; availability across master store and sub-stores; asset BOM/where-used; technician part requests and storekeeper issue/return as Stock Entries; inter-store transfers (Material Transfer via Transit); outbox, idempotency and exceptions queue; Admin approval inbox; MR (Purchase) for shortages with PO/receipt tracking; repair location on WOs; critical-spares dashboard; onboarding templates for spares/BOM; time + meter PM; checklist library v1; calendar; in-app/push/email notifications; basic escalation; offline technician PWA; core dashboards (PM compliance, MTBF, MTTR, downtime, backlog)	A pilot area at MP uses the system for all breakdowns and PMs for 4 weeks; ≥ 90% of breakdowns captured via QR; failure-code data quality reviewed
2. Roll-out to the Ancillary Plant and Campus	6–8 weeks	AP and CP configuration (calendars, storerooms, teams, escalation contacts); 3-site comparison dashboard; cross-site stock visibility; PM templates applied to asset groups across sites; TPM CIL operator checklists; job card printing; data-quality fixes from the pilot	All three sites live; management uses the 3-site dashboard
3. Standardisation and control	6–8 weeks	Checklist versioning and approval; PM kitting via the Kitting warehouse; sub-store replenishment (Item Reorder type Transfer); reorder-level proposals pushed to ERPNext Item Reorder after Admin approval; workshop repair jobs and queues (Mechanical/Electrical shop); repairable-spares	PM compliance and planned-work % reported monthly per site; ≥ 95% of maintenance stock movements originate from CMMS WOs

Phase	Duration (indicative)	Scope	Exit criteria
		cycle; external repair Option R1 (R2 if required); spares consumption analytics; capacity planning; cost roll-up; RCA records; repeat-failure flags; WhatsApp/SMS for P1	
4. Reliability and integration	Ongoing	Condition-based triggers; PLC/SCADA/IoT meter ingestion; OEE (with production data); schedule compliance; BI access; rotables; certifications; 5S audits; predictive analytics on accumulated failure data	Agreed per business case

Rollout practice (from research): pilot first, then expand, as eMaint recommends [S16]. MaintainX says its average customer takes about three weeks to implement one site [S1]. Treat this as a rough benchmark for data loading and training per site, not a commitment.

Early wins to plan for:

- A clean asset master for all three sites, with QR labels.
- A failure-code list per major asset class, agreed with senior technicians before go-live.
- A first **maintenance spares list** created in ERPNext (greenfield), with criticality flagged for parts of A-critical machines, and opening stock posted per store (master and sub-stores) by stores.
- 10–20 standard PM checklists for the most critical machine classes, shared across sites.

11. Open questions for the user

Answered and removed:

- v0.2: sites and machine count; devices (technicians phone-first; engineers mobile + desktop); hosting (on-premise, PyCharm → Ansible); existing data (greenfield); ERP/accounting (ERPNext for POs and inventory; Tally out of scope).
- v0.3: ERPNext version (**v16**) and test instance (**provided**); master-data sync direction (ERPNext → CMMS, read-mostly); store structure (**master store + a few sub-stores**); repair locations (**in-situ, Mechanical shop, Electrical shop, external**); who issues (**storekeeper; technicians request only**); system of record (**Option A, no CMMS stock ledger**); approver for maintenance MRs/POs (**Admin**).
- v0.4: Admin model (**exactly one Admin per instance, single approver across all sites**; per-site admin, second approver and value threshold are not needed; leave-cover delegate is optional).

Still open (top five first):

1. **How stores book issues today:** Is the current "material issue voucher" an ERPNext Stock Entry with purpose **Material Issue**, or a **Material Transfer** to a consumption/department warehouse? If it is a transfer, which warehouse, and how and when is it expensed? This decides the issue mapping mode (§9.4.1). Also: which cost centre and expense account should maintenance consumption use, per site, department or asset class?
2. **ERPNext v16 settings:** Is perpetual inventory enabled? Is negative stock allowed (globally or per item)? Is Item auto-reorder already in use? Is Stock Reservation / Auto Reserve Stock enabled for sales or production (it reduces what maintenance can use)? Where is ERPNext hosted relative to the CMMS server?
3. **Stores in detail:** Names and locations of the master store and each sub-store, and which sites/areas each serves. Current warehouse tree, item naming series and item groups for spares. Can we add the Kitting, Repair Quarantine, Mechanical/Electrical Shop WIP, At Vendor for Repair and Maintenance Transit warehouses?
4. **External repair and GST:** How are goods sent out for repair documented today (returnable delivery challan, e-Waybill, job-work/ITC-04)? Is the India Compliance app installed? This decides Option R1 (transfer + service PO) vs R2 (ERPNext Subcontracting).

5. **Repairables accounting:** How should finance value a failed unit received into quarantine, a repaired unit returned to stock (is the repair cost capitalised or expensed?) and scrap? Does the Admin have an ERPNext user for PO approval, and who is the optional leave-cover delegate (if any)?
6. **Workshops:** Where are the Mechanical and Electrical shops (which site)? Do they repair for all three sites? Are workshop technicians a separate team?
7. **Machine split and site roles:** Roughly how many of the ~500 machines are at each site, and what does each site produce or host?
8. **Plant areas, machine classes and criticality:** Which process areas and main machine types exist, and which are A-critical?
9. **Hierarchy depth:** Machine level only in the MVP, or key components too (motors, gearboxes, hydraulic units)?
10. **Users and shifts:** How many people are in each role per site, and how many shifts? Are storekeepers shared between stores?
11. **Phones:** Company-issued or personal (affects certificate trust, MDM, data-protection notices)? Minimum Android version? Will operators have phones, or is a shared kiosk needed?
12. **Network:** Wi-Fi/4G coverage inside each site (kiln and press areas); is a site-to-site VPN/leased link already in place? Where will the on-prem server live, and is there a second site for backup copies?
13. **Domain and certificates:** Does the company own a domain with DNS API access (for ACME DNS-01), or is an internal CA preferred?
14. **CI:** Is there a Git server/runner for an automated test stage (including ERPNext contract tests) before the Ansible deploy? Docker acceptable, or native systemd services only?
15. **Production data:** Is production count, cycle time or running hours available (manual, PLC, MES) for OEE and meter-based PM?
16. **Safety:** Are LOTO/permit-to-work procedures in place that the CMMS should enforce?
17. **TPM/5S maturity:** Is there an existing TPM/5S programme to digitise?
18. **WO closure approvals:** Who verifies WO completion at each site, and for which WO types/priorities?
19. **Language:** Should master data be kept in both Tamil and English from day one?
20. **Notifications:** Is WhatsApp acceptable for breakdown alerts?
21. **Compliance/audit:** Do any ISO or statutory inspections require specific records or e-signatures?
22. **KPI targets:** Which 3–5 KPIs matter most in year one?
23. **Team and timeline:** Team size and target pilot date? Is the Main Plant the right pilot site?
24. **Retention:** How long should maintenance history and audit logs be kept?

12. Sources

Research carried out 12 Oct 2026 (v0.1) and extended for ERPNext, deployment and PWA/TLS topics (v0.2). Vendor pricing and tiers are as published on that date and may change. “Secondary” marks third-party sources used to cross-check vendor pages.

MaintainX

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    - Mobile vs. Web: https://help.getmaintainx.com/mobile-vs-web-overview
    - Create a Work Request: https://help.getmaintainx.com/create-a-work-request
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